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ABBREVIATIONS

Abbreviation	Definition
AACCS	Action Against CCS (community campaign organisation)
AGI	Above-Ground Installation
BEIS	Department for Business, Energy and Industrial Strategy (now DESNZ)
CCC	Climate Change Committee
CCS	Carbon Capture and Storage
CDA	Critical Discourse Analysis
CfD	Contract for Difference
CO_2	Carbon dioxide
DCO	Development Consent Order
DESNZ	Department for Energy Security and Net Zero
EIA	Environmental Impact Assessment
ESRC	Economic and Social Research Council
ETS	Emissions Trading Scheme
GBP	Pounds Sterling (£)
GCCA	Global Cement and Concrete Association
GJ	Gigajoule
GOV.UK	UK Government official web portal
ICC	Industrial Carbon Capture (DESNZ business model)
IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
LC3	Limestone Calcined Clay Cement
LCO_2A	Levelised Cost of CO_2 Avoided (£ or USD per tonne CO_2)
MAC	Marginal Abatement Cost
MEA	Monoethanolamine (post-combustion capture solvent)
MPA	Mineral Products Association
Mt	Megatonne (10^6 metric tonnes)
NPS EN-1	National Policy Statement for Energy Infrastructure (overarching)

NPS EN-4	National Policy Statement for Gas Supply Infrastructure and Pipelines
NSIP	Nationally Significant Infrastructure Project
NVivo	Qualitative data analysis software (Lumivero)
NWF	National Wealth Fund
PRTR	Pollutant Release and Transfer Register
SCM	Supplementary Cementitious Material
SLO	Social License to Operate
t CO_2	Tonne of carbon dioxide
TEA	Techno-Economic Analysis
UK	United Kingdom
UK-PRTR	UK Pollutant Release and Transfer Register
USD	United States Dollar ($)
VADER	Valence Aware Dictionary and Sentiment Reasoner
ZEP	Zero Emissions Platform (EU CCS industry coalition)

ABSTRACT

As the world pursues efforts to decarbonise its hard-to-abate sectors, the Peak Cluster-Morecambe project represents one of the most ambitious CCS initiatives in the cement industry. With a proposal to transport CO2 emissions from anchor cement and lime facilities in the UK to depleted gas reservoirs, this research evaluates the policy, economic viability and public perception of this project using a convergent parallel mixed-methods case study design, combining critical document analysis, techno-economic assessment and a discourse analysis of social licence to operate.

The analysis of government and industry documents using a lock-in framework reveals a strong path-dependent policy environment with a mean lock-in score of 4.4 out of 7, with limited allowance for alternative pathways. The techno-economic component employs a Monte Carlo simulation to assess the levelized cost of CO_2 avoidance, with results indicating a median cost of USD 131.4 per tonne, which substantially exceeds the 2026 UK ETS reference price of USD 65.2, implying a long-term subsidy requirement to sustain CCS. Comparative marginal abatement cost analysis further demonstrates substitute decarbonisation options that could achieve meaningful emission reduction at a lower cost.

Discourse analysis of 191 social media records indicates that the project has not secured a social license to operate beyond the conditional holding and scrutiny phase. Public concerns are framed around place identity and landscape protection, procedural fairness, and environmental and safety risks, resulting in widespread distrust of project developers and regulatory bodies.

Integrating these findings, the study uncovers a generative mechanism where path-dependent policy regimes advance CCS deployment while simultaneously undermining social legitimacy. This study concludes that despite institutional lock-in in decarbonisation pathways in the UK cement industry, opportunities remain for procedural reform, stakeholder engagement and pluralistic approaches to achieving net-zero targets.

Keywords: *carbon capture and storage; carbon lock-in; cement decarbonisation; path dependency; techno-economic analysis; Peak Cluster*

DECLARATION

I hereby declare that this dissertation is the student's original work and that no portion of the work referred to in the dissertation has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Alhassan Abdulai

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CHAPTER ONE

INTRODUCTION

1.0 Background

Global CO_2 concentration levels have risen by more than 30% post-industrial revolution, with the Intergovernmental Panel on Climate Change (IPCC) emphasizing substantial reductions in carbon dioxide (CO_2) across all sectors to limit global temperature rise to 1.5°C, in line with the Paris Agreement (IPCC, 2023). The cement industry contributes a significant share (7-8%) of global anthropogenic CO_2 emissions, largely due to calcination and clinker formation (Li and Unluer, 2025). This makes the cement industry particularly hard to abate, with numerous technological and policy considerations being proposed to decarbonize the sector such as switching to renewable energy sources, developing low-carbon cements, and implementing carbon capture and storage technologies (IEA, 2023).

The United Kingdom's (UK) cement industry collectively produces around 10 million tonnes of Portland cement (PC), contributing nearly a billion pounds annually to the economy (OEC, 2024; Mineral Products Association, 2025). Currently, the UK cement and concrete industry accounts for about 1.5% of national greenhouse gas (GHG) emissions and is pursuing ambitious targets to reach net zero by 2050, with a roadmap in place (Mineral Products Association, 2025). Decarbonisation options such as electrification, cleaner transport, fuel switching, and clinker reduction in construction have the potential to cut emissions by 39%, necessitating the development of deep decarbonisation technologies like low-carbon cements and carbon capture and storage (CCS) (Sherif *et al.*, 2025).

CCS technologies capture CO_2 at emission points (source) and store the captured CO_2 through high-pressure injection into deep geological formations such as depleted oil and gas reservoirs and saline aquifers. The main appeal of CCS is its versatility, which enables it to be combined with other technologies like bioenergy or direct air capture to achieve negative emissions by decarbonising existing infrastructure (Fuss *et al.*, 2018). In the UK, CCS projects are being developed through industrial cluster strategies, including the Peak Cluster project, which aims to significantly decarbonise cement production across multiple locations in the Peak District and neighbouring counties, such as Derbyshire and Staffordshire, where 40% of the UK's lime and cement is produced. Led by Progressive Energy and involving major producers like Tarmac, Breedon, Holcim, and Buxton Lime (SigmaRoc), the project targets a reduction of 3 million tonnes of CO_2 annually. It proposes a 200 km underground pipeline to transport CO_2 through Staffordshire and Cheshire to the Wirral coast, where it will be stored permanently in depleted gas reservoirs (Peak Cluster, 2026).

1.1 The Problem Statement

The cement industry's ambition to decarbonise its processes has led to increased promotion of CCS as a necessary mitigation strategy (IEA, 2023). This aspiration has, however, been heavily contested, with critics expressing concerns over the high cost of CCS infrastructure, technological uncertainties, and overreliance on government subsidies and policy support to enable projects to thrive (Gallego Dávila and Aagesen, 2024; IEA, 2025). Deployment of large-scale CCS projects has sparked major debate, with concerns about their economic viability, environmental impact, and the uncertain role of CCS in long-term decarbonization due to its propensity to introduce technological lock-in (Erickson *et al.*, 2015; Twidale, 2025).

However, this pathway is heavily contested as alternative low-carbon cement technologies already exist and could potentially be more cost-effective mitigation options (IEA, 2023; Sherif *et al.*, 2025). Emerging start-ups like TerraCO2, Sublime and Brimstone are developing alternative low-carbon cement technologies that could be more cost effective mitigation options (Ellis, 2023; Brimstone, 2026; Terra CO2, 2026). Although low-carbon cement shows some potential, recent studies underscore CCS as the most promising technology for decarbonising the cement industry, with the Global Cement and Concrete Association (GCCA) estimating 36% total carbon savings by 2050 through CCS, making it the leading mitigation option (GCCA, 2023).

This has prompted questions about the opportunity cost of CCS projects in the cement industry (IEA, 2023; He *et al.*, 2025), and existing debates are further complicated by emerging concerns over public acceptance of CCS infrastructure, which could affect the social feasibility of large-scale CCS projects (Niranjan, 2024). Within the UK, the Peak Cluster project exemplifies these tensions, necessitating public scrutiny and policy support. While existing research has examined the technical, economic, and policy perspectives of CCS projects, a critical gap exists: a comprehensive analysis that simultaneously considers policy incentives, opportunity costs, and public perception within a specific context remains limited. Without a holistic assessment, the UK risks a "path-dependent" lock-in that ignores public acceptance of CCS infrastructure and social feasibility.

1.2 Research Aim and Objectives

This study aims to critically evaluate the policy drivers, economic viability, and public perception of CCS in the UK cement sector, using the Peak Cluster Project as a case study, with particular attention to its opportunity cost relative to alternative decarbonization pathways.

The study will pursue the following specific objectives:

1. To critically analyse UK CCS policy and regulatory frameworks and how they create institutional path dependency.
2. To evaluate the economic viability and opportunity costs of CCS in the context of infrastructure lock-in.
3. To examine public perception and social acceptance of CCS in the UK, focusing on stakeholder concerns related to the Peak Cluster project.

These objectives will attempt to answer the following questions:

1. To what extent is CCS an economically viable and socially acceptable decarbonization strategy for the UK cement industry and does it create a technological "lock-in"?
2. How does CCS compare with alternative decarbonization pathways in terms of long-term costs?

3. What are the main public concerns regarding the project that might challenge its social “path"?

1.3 Structure of the Thesis

Chapter 1 will introduce the research, background, aims, and objectives. Chapter 2 will focus on a literature review of cement decarbonisation, CCS, and public perception. Chapter 3 will describe the study area and outline the research methodology. Chapter 4 presents the study's findings, and Chapter 5 concludes the study and offers recommendations.

CHAPTER TWO

LITERATURE REVIEW

2.1 Understanding Path Dependency and Carbon Lock-in

Path dependency is best described as the dependence of future societal decision processes or ecological outcomes on those that have occurred in the past, thereby influencing adaptation (Ernst and Preston, 2017). A specialised form of path dependence is carbon lock-in because the interacting technological, institutional and behavioural systems, by design, entrench fossil-based trajectories. This framing of the techno-institutional complexities and the relationship between technology and governance that sustains carbon regimes has been discussed widely in literature (Buschmann and Oels, 2019; Kotilainen *et al.*, 2019).

Lock-ins are however not merely technical but rather discursive and political and as such, discursive turning points in lock-ins can shape transitions and serve as levers that demonstrate how a dominant narrative can disrupt or sustain a carbon pathway (Buschmann and Oels, 2019). Policy discourses and advocacy can also reinforce lock-in or unlock new pathways that drive the shift towards decarbonization through strategic discourse and research and development funding choices (Robertson, 2022). The long life span of energy-using assets like blast furnaces in the steel industry is a salient reason for carbon lock-in and has been a central impediment to rapid decarbonization of the steel industry (Vogl, Olsson and Nykvist, 2021).

Literature on lock-in discusses three channels and mechanisms that reinforce lock-in, namely infrastructural/technological, institutional and behavioural. Infrastructural lock-in is primarily reinforced by the capital-intensive nature of plants such as blast furnaces, creating sunk costs and long investment horizons, ultimately hindering the switch to low-carbon technologies. Literature discussing the decarbonization of such industries has called for a phased asset retirement, while recognising the path-dependent nature of such transitions, especially in energy-intensive processing industries (Wesseling *et al.*, 2017; Vogl, Olsson and Nykvist, 2021). Institutional carbon lock-in, however, is reinforced by government structures, regulations and market institutions, with policy windows and political opportunities capable of loosening or reinforcing lock-in (Seto *et al.*, 2016). Evidence for this can be extended to policy mix found at the intersection of transportation and energy regimes like the Nordic electric mobility (Kotilainen *et al.*, 2019). The third channel which is behavioural lock-in looks at the demand-side habits, norms and expectations which dictate consumer preferences and organisational routines which contribute to path dependency (Biddau, Brondi and Cottone, 2022). These channels, however, are influenced by a discursive political economy dimension which constitute an interplay of non-state and sub-national actors playing a critical role in breaking or maintaining lock-in through normalisation, capacity building and coalition formation to influence lock-in dynamics (Van Der Ven, Bernstein and Hoffmann, 2017; Bernstein and Hoffmann, 2018).

Path creation debates have also emerged within the path dependency debate, arguing that industrial regions can leverage existing assets to build new low-carbon technologies, as evidenced by net-zero industrial clusters (Lai and Devine-Wright, 2024). This argument illustrates the techno-institutional complexities surrounding lock-in and why decoupling these two could reinforce lock-in (Hoicka *et al.*, 2021). Hence, to disrupt lock-in, transition planning, regulatory and market frictions, and a coordinated phase-in/phase-out approach is required through multidimensional research to address the constraints in the decarbonization roadmap (Grubert and Hastings-Simon, 2022).

Political momentum is a very useful tool in disrupting carbon lock-in because it helps to normalise decarbonisation practices through the building of coalition across scales (national, subnational, corporate). Central levers for disruption, as evidenced in previous bodies of work, include normalisation, capacity building and coalition building in the form of policy mixes (Bernstein and Hoffmann, 2018; Kotilainen *et al.*, 2019). The role of subnational and nonstate actors as transformative agents catalysing multi-level disruptions serves as an incentive and support structure for breaking lock-in (Van Der Ven, Bernstein and Hoffmann, 2017). Discursive shifts also play multiple roles in this process, including shaping narratives, framing of decarbonisation as both economically and socially beneficial, thereby creating “discursive turning points” that reframe assets as transitional liabilities rather than long-term investments (Bernstein and Hoffmann, 2018; Buschmann and Oels, 2019). As a result, there is a need for explicit mid-transition and phase-out planning to avoid misalignments between policy promises and industrial capabilities.

Decarbonisation is constrained by infrastructural, institutional and behavioural dimensions and can be overcome through integrated strategies that rewire institutions, reframe the narratives to mobilise political and financial support, empower subnational and non-state actors, and tailor interventions to place-based sociotechnical configurations, thereby breaking carbon lock-ins and path dependencies.

2.2 Institutional Path Dependency and the Institutional Carbon Capture Model

The pursuit of CCS and industrial decarbonisation by the UK government is influenced by a mix of policy ambitions, industry finance, and complex path-dependent dynamics. The framing of policies, regulatory structures and investment incentives in a manner that became constraints for later transitions to lower-carbon industrial configurations is attributed to carbon lock-in and socio-technical path dependence in numerous studies (Erickson *et al.*, 2015; Barazza and Strachan, 2020). As such, business models, investment decisions, return horizons and market eligibility for CCS-enabled industrial services are shaped by perceived policy stability, the sequencing of policy instruments and the incumbent technologies and infrastructure (Foxon *et al.*, 2013; Erickson *et al.*, 2015).

A critical precursor necessary to sustain long-term investment in low-carbon technologies is policy stability. Lasting policy frameworks that are not prone to reversals are key in boosting investor confidence when addressing problems like climate change (Rietig and Laing, 2017). In parliamentary systems like the UK, consensus across parties and multi-level governance is needed to provide investments in low-carbon technologies (Rietig and Laing, 2017). The UK’s policy narratives on decarbonisation pathways, such as CCS and bioenergy, tend to rely on technocratic imaginaries that mobilise co-investments and industrial co-financing to become viable. State-industry interactions in the biomass and bioenergy sector have been shaped by policy, leading to the creation of path-dependent expectations that reinforce incumbent infrastructure, with the future benefits of new pathways being projected and not adopted early enough (Levidow and Papaioannou, 2016). This is a reality which resonates with most CCS business models because policy signals (subsidies, contracts for difference, capacity payment track records, loan guarantees) are the backbone for long-term CCS asset deployment within industrial clusters (Levidow and Papaioannou, 2016; Barazza and Strachan, 2020).

Regulatory entrenchment, capital expenditure and scale of projects have been identified as methods of assessing lock-in strength which explains why coal, gas and oil based assets exhibit the highest risk (Erickson *et al.*, 2015). Policy windows can also catalyse new industrial paths as demonstrated in offshore wind projects (Dawley, 2014). However, some studies also warn strongly against overreliance on disruptive narratives stating that it could mislead policy and research unless there is a complementarity between continuity and change as evidenced in CCS projects in the energy sector (Winskel, 2018).

CCS in the UK is advanced through industrial cluster approaches co-locating capture, transport and storage infrastructure at energy-intensive industries in order to realise scale and share cost (Dawley, 2014; Lai and Devine-Wright, 2024). Some studies have also emphasized the urgency of national strategies like carbon pricing, subsidies and capacity mechanisms coupled with local capabilities and institutional arrangements to maintain such clusters (Barazza and Strachan, 2020). Narrative-based policy can also shape innovation priorities by favouring or counteracting innovation priorities and as such, policy design should incorporate diverse evidence sources (industry, local communities, technical experts) in order to avoid lock-ins that privilege established systems (Levidow and Papaioannou, 2016). CCS social acceptance varies based on deployment context, risk perceptions, and governance arrangements. Witte (2021) emphasised the importance of incorporating social acceptance considerations into policy to avoid the lock-in of unsustainable practices and foster just transitions, especially for CCS business models that require a social license to operate, regulatory clarity and transparent risk governance (Witte, 2021).

Given the lock-in risk in CCS deployment, a phased deployment with policy signals that outlast electoral cycles and risk sharing across industry and government to create room for early projects to learn and reduce cost without foreclosing alternative decarbonisation pathways (Foxon *et al.*, 2013). UK's CCS deployment is leveraging the existing industrial ecosystem and logistic advantages. However, efforts to avoid premature lock-in to particular technologies are necessary and should incorporate mechanisms for reconfiguration in response to changes in technology costs and regulations (Berg and Hassink, 2014). Policies should foster flexibility by balancing near-term deployment with associated investment in research, innovation, and modularity, which permits substitution of technology or reconfiguration in the wake of cheaper or safer options (Río and Labandeira, 2009; Rietig and Laing, 2017). Inclusive deliberation incorporating the concerns of industry incumbents, suppliers, communities, environmental groups should guide the decision of policymakers in order to mitigate narrative-driven lock-ins (Witte, 2021).

2.3 Hardware Lock-in of Pipeline Infrastructure in CCS

Pipeline infrastructure for the transport of CO_2 across multiple analyses has been found to exhibit substantial economies of scale with pipelines of larger-diameter and higher volume reducing transport cost per tonne (Skobelev, Cherepovitsyna and Guseva, 2023). The economies of scale reduce unit transport cost of CO_2 with higher utilisation, while clustering emission sources into hubs improves cost efficiency by amortising pipeline and CO_2 compression assets across multiple emitters (Breunig *et al.*, 2022). CCS hubs and clustered industrial networks therefore target scale economics in CCS deployment by sharing pipelines, compression, monitoring and storage infrastructure across multiple hard-to-abate industries (cement, power, iron/steel, chemicals) (Hammond, 2022; Wang *et al.*, 2024; He *et al.*, 2025; Liu *et al.*, 2025). Comparative regional assessments in areas such as Europe and China equally indicated that emission aggregation over pipeline networks utilizing offshore sinks or proximate storage sinks improve economics substantially (Wang *et al.*, 2024). The linking of plants to shared pipelines and storage sites reduces costs relative to standalone capture and transport capture and transport options (Wang *et al.*, 2024).

Based on multiple studies, the benefits of a shared infrastructure for CCS deployment become pronounced as deployment scales from hundreds to thousands of MtCO2/year, because costs decline due to better utilisation of shared infrastructure, equipment and monitoring. (Zhang *et al.*, 2024). CCS deployment costs were also found to be sensitive to the chosen capture technology (post-combustion, oxy-fuel, CCS coupling) and policy incentives (Ziobrowski and Rotkegel, 2024). Overall, broader CCS hub designs which include multiple emission sources (cement, power, steel, chemicals) was found to enable higher throughputs and asset utilisation and lower unit costs (Zhang *et al.*, 2024). As a result, policy frameworks have consistently advocated for shared transport corridors and clustered storage models in CCS deployment (Hammond, 2022; OECD, 2022).

The CCS infrastructure needs to be paired with strong governance, stable financing, and robust public-private collaboration inorder to realise success in CCS deployment (Safari, Roy and Assadi, 2021). When the appropriate policy and capacity exist, deployment options (cluster vs. solitary) determine whether economies of scale can be realised. Experiences from national and regional deployment of CCS across the UK, Europe and China show that the success of hubs in attaining economies of scale through shared pipelines, and cross-border or cross-sector storage will only materialise if there is a sufficient market demand, pipeline rights-of-way, and regulatory clarity (Safari, Roy and Assadi, 2021; OECD, 2022).

2.4 Economic Viability and Subsidy Considerations for CCS

While CCS/CCS has been repeatedly cited as a potential pathway to decarbonising the cement industry, literature continuously identifies challenges such as high capital cost, energy penalties for capture and dependence on public policy to reach cost-competitive levels (Zhou, Pan and Mao, 2023; Ampomah *et al.*, 2024). As such, numerous studies have repeatedly identified subsidies and carbon pricing as pivotal instruments central to unlocking CCS deployment in cement and hard -to-abate industries ((Stigson, Hansson and Lind, 2012; Faber *et al.*, 2022; Zoback and Smit, 2023).

The cost drivers and current CCS economics indicate that capture costs dominate total CCS costs for cement plants, with industry-level Techno-Economic Assessment (TEA) and reviews showing that capture and transport costs form the largest share of the cost stack, storage costs are significant yet smaller than capture costs, and energy penalties increase total energy use when capture is retrofitted to cement plants (Coninck and Benson, 2014; Ampomah et al., 2024). Estimates indicate that capture costs can range from EUR 42.4 to 83.5 per tCO_2 avoided for cement, serving as a major barrier to economic viability in the absence of policy support and high carbon prices (Strunge et al., 2024). These findings align with broader cement CCS TEA syntheses, which show an economic model that is capital expenditure (CAPEX)- and operational expenditure (OPEX)- intensive, with substantial uncertainties in the absence of subsidies and high carbon prices (Faber et al., 2022).

Energy penalty on CO_2 capture contribute to cost increment of between 3 – 10% in energy use per tonne of clinker formation in some assessments, with higher costs in some instances depending on the technology deployed and scale (Friedl *et al.*, 2023). While cement-specific capture is relatively favourable, the residual energy and processing penalties still raise the levelized cost of cement production if CCS is considered (Coninck and Benson, 2014; Friedl *et al.*, 2023).

A comparative cost of CCS vs alternative decarbonisation in some studies has found CCS to be cost-competitive in only when there exist high carbon prices and favourable policies, while other alternatives such as cement alternatives, CCU routes like mineralisation maybe more cost competitive in certain geographies or supply chains (Zhou, Pan and Mao, 2023; Esiri, Jambol and Ozowe, 2024). In geographies where tax credits, direct subsidies, loan guarantees, and public investment in shared infrastructure exist, CCS deployment becomes more viable financially because CAPEX and OPEX, as well as perceived risks, are reduced relative to business-as-usual production processes (Stigson, Hansson and Lind, 2012; Faber *et al.*, 2022; Ampomah *et al.*, 2024). While carbon prices alter the economics of CCS deployment, in certain contexts, under plausible carbon prices, there is still a need for favourable policies and subsidies for large-scale uptake indicating that subsidies are not ancillary but central to the deployment of CCS in the cement sector (Friedl *et al.*, 2023; Zoback and Smit, 2023)..

2.5 The Opportunity Cost of Alternative Binders

Calcinated clay and limestone-calcined clay cement (LC3), alongside other alternative binders and alkali-activated materials, have been identified as promising strategies for reducing clinker demand, lowering CO_2 emissions and realising cost savings on energy. The opportunity cost of sticking with high-clinker conventional Portland cement (PC) includes higher CO_2 emissions and higher materials cost (Scrivener, John and Gartner, 2018; Mishra *et al.*, 2022; Isaksson *et al.*, 2023). Studies on the quantification of CO_2 reductions via LC3 demonstrated meaningful CO_2 reduction with the usage of relatively inexpensive locally available inputs (Scrivener, John and Gartner, 2018). The regional availability of the alternative binders is key to achieving these reductions making the opportunity cost of exploring this option context-specific (Gartner and Hirao, 2015).

LC3 and calcined clay blends are reported to reduce CO_2 per ton of cement by between 20-40% relative to PC, depending on the composition (LC3-50 or LC3 variants)(Ram *et al.*, 2022; Villagrán-Zaccardi *et al.*, 2022). Further analysis of LC3 composition indicated that it could match PC in strength and still deliver on lower carbon emission and clinker replacement (Mishra *et al.*, 2022).

The policy and procurement mechanisms of materials in different contexts influence opportunity costs. As such, updated standards could accelerate the uptake of alternative binders, increasing the opportunity cost savings for adoption (Serdar *et al.*, 2019; Busch *et al.*, 2025). Gaps in the standards of testing have obscured the true performance and cost benefits. For many agricultural ashes and calcinated clays, it has been realised that standard mortar testing might not fully capture concrete and block performance which leads to underestimation of cost savings (Gartner and Hirao, 2015). CCS economics often elevate the attractiveness of calcinated clays and LC3 because CCS remains expensive and thus constitutes a higher opportunity cost in comparison to low CO_2 alternatives (Scrivener, John and Gartner, 2018; Khan *et al.*, 2023; Tau *et al.*, 2023).

Across Europe, Latin America and Sub-Saharan Africa (SSA), different opportunity costs are exhibited primarily due to local clay resources available, calcination energy costs and cement plant configurations, with opportunity costs being more favourable if calcinated clay is close to the production site (Scrivener, John and Gartner, 2018; Tau *et al.*, 2023). Striving for synergies between LC3 and other low carbon strategies like alternative fuels, fly ash, can increase avoided clinker demand (Bhattacherjee *et al.*, 2021).

Current debate on LC3 revolves around the scale of deployment required to displace clinker globally, with many studies reporting a strong potential deployed incrementally through improved supply chain and market incentives (Mishra *et al.*, 2022; Villagrán-Zaccardi *et al.*, 2022). The role of CCS as a major technology for emission reduction has therefore lost some of its appeal due to the cost and energy penalties, thereby raising the appeal of SCMs like LC3 (Gartner and Hirao, 2015; Khan *et al.*, 2023).

2.5 Social Path-Breakers as a Catalyst for Change

Social licence to operate (SLO) can be defined as the "ongoing acceptance or approval of a development by local communities or stakeholders, grounded in legitimacy, credibility and trust" in the project and its actors (Thomson and Boutilier, 2011). SLO has been used to explain why some CCS projects stall and how corporate and project-level legitimacy lead to major failures or broader social acceptance (Jijelava and Vanclay, 2017; Gough and Mander, 2019). For SLO to be incorporated into CCS deployment designs, it must be understood that CCS is multi-layered, covering initial legitimacy to trust-based approval, and in some models, psychological identification with the project. As a result, researchers have explained why "transfer" of SLO from one CCS project to another is might be inappropriate if local histories, governance relations and stakeholders are not considered (Colton *et al.*, 2016; Jijelava and Vanclay, 2017).

Studies on public perception on CCS indicate that the understanding of risks associated with CCS is highly contextual, and is generally shaped by controllability, trust in institutions and perceived safety. Analysis by Stalker *et al.,* (2021) and Gordon *et al.,* indicate that acceptance levels of CCS from analogous subsurface technologies by communities were affected by risk communication, demonstrable regulatory monitoring, track record of safe and benefits for communities, and governance arrangements (Stalker *et al.*, 2022; Gordon, Balta-Ozkan and Nabavi, 2023). A further review of CCS adjacent technologies indicates that technology deployment will also need to incorporate issues of fairness, transparency in decision making and trust in regulators and operators since deployment of these technologies are over a long time horizon and sustainability of the SLO is key (Arpaillange, 2017; Gough and Mander, 2019).

The social impact assessment (SIA) and SLO are key in offshore CCS deployments because community-government-marine relationships add an extra layer of complexity. The Tomakomai CCS project demonstrated how the public responds to project deployment based on the historical and coastal cultural relationships they have had with the sea, engagement on monitoring and perception of local benefits and risks (Gough and Mander, 2019). The legal and regulatory analyses of CO_2 storage and pipeline permits indicated that these measures contributed to a broader governance environment within which SLO evolved (Roggenkamp, 2020)

It is also worth noting that major lessons can be learned from historical controversies surrounding CCS project deployments and communications failures. A typical example is the case of the Barendrecht CO_2 storage project, where the perception of being forced to accept CCS led to strong outrage and ultimately project termination. This further reinforces the strategic importance of community-led engagement, perceived consent and early robust risk communication to strengthen SLO (Dolan, 2020). An analysis of BP’s strategy on pipelines and energy infrastructure demonstrates that legitimacy, credibility and trust levels enhance SLO in CCS pipelines if conditions such as stakeholder engagement, transparent governance and demonstrable safety records are met (Jijelava and Vanclay, 2017).

Evidence from some studies have indicated that framing CCS as part of an equitable climate solution with robust governance and transparent information supports SLO and public acceptance in the long term (Gough and Mander, 2019; Gordon, Balta-Ozkan and Nabavi, 2023). Literature on public acceptance notes that pipeline repurposing and storage decisions require careful site-specific engagement to strengthen SLO as long as safety, environmental and equity concerns are addressed (Grecksch, 2021).

So, while the importance of SLO to operate cannot be overemphasized, it is worth noting that substantial debate still exists on the exact mechanism required to transfer SLO across projects. While some studies argue that SLO is project-specific and context dependent, others argue for attributions for legitimacy or trust, which is relevant across multiple projects (Stalker *et al.*, 2022). There is also a lot of discussion on whether public acceptance should be framed primarily as “acceptance” or a broader set of qualitative and legitimacy concerns with most authors proposing a multidimensional view rather than a single acceptance metric (Colton *et al.*, 2016; Jijelava and Vanclay, 2017)

2.6 Synthesis and Identified Research Gap

The literature review converged on three propositions. Firstly, industrial decarbonisation is governed by path-dependent self-reinforcing dynamics, which cause infrastructural, institutional and behavioural lock-ins to co-evolve into incumbent trajectories (Erickson *et al.*, 2015; Seto *et al.*, 2016; Buschmann and Oels, 2019). Secondly, the CCS economics in cement remain marginal at the current carbon prices, and are shaped by the design of policy instruments and not just technological costs alone (Faber *et al.*, 2022; Friedl *et al.*, 2023; Strunge *et al.*, 2024). Thirdly, the social feasibility of a CCS infrastructure will continue to evolve as a renegotiated SLO, an absence or loss of which can stall a technically viable project (Jijelava and Vanclay, 2017; Dolan, 2020; Witte, 2021).

These strands are all real concerns that have each been isolated and treated as a standalone in literature, and have rarely been brought together in a single empirical frame, and rarely within the cement-sector case in the United Kingdom. Path dependency literature is often conceptual and focused on energy systems studies at the national scale, which has left sub-sectoral institutional infrastructure least examined. Techno-economic literature has primarily viewed and examined CCS in the cement industry as a cost-engineering problem, which tends to treat policy and subsidies as an external parameter rather than an internal expression of institutional commitment. SLO literature focused more on legitimacy, credibility and trust in a post-implementation project-specific context. SLO literature had little engagement with policy and economic levers that shape the discourse in the first place. As such, the literature presented no integrated account of how the institutional architecture, economic case and social license interact to create or fail to produce a flexible decarbonisation pathway at a single industrial cluster.

This gap has consequences for the UK cement sector, especially the Peak Cluster – Morecambe Net Zero project, which is being advanced through CCS policy instruments at a critical moment when alternative low carbon binders could become cost-effective substitutes. It also comes at a time when the project is being contested in its consenting corridor. The question of whether this commitment to the project will constitute an optimal decarbonisation pathway, an avoidable lock-in or both remains unanswered in any literature in isolation.

This current study addresses this gap by analysing these three parallel dimensions within a single case. Objective 1 will examine institutional lock-in architecture in and its signature, objective 2 will quantify the economic case and its opportunity cost against alternatives, and objective 3 will reconstruct the SLO and its associated frames at the pre-deployment stage. By integrating these objectives, the study will contribute an empirically grounded account of how path-dependent decarbonisation is produced and the conditions under which it might be broken.

CHAPTER THREE

3.1 Research Methodology

This chapter outlines the methodological framework used to examine the role of CCS in the UK cement sector across its policy, economic, and social dimensions. The framing of industrial decarbonisation in chapter 2 indicated that it is a socio-technical, path-dependent and politically contested process (Seto *et al.*, 2016; Buschmann and Oels, 2019). As a result, the study requires a design that can work simultaneously across institutional, techno-economic and discursive domains. A convergent mixed methods case-study design, grounded in critical realism, meets this requirement.

3.2 Research Philosophy

The critical-realist paradigm (Bhaskar, 2008; Maxwell, 2011; Fletcher, 2016) is adopted for this study. Critical realism's stratified ontology enables research to differentiate the actual, empirical and real, which is a generative mechanism that has been inferred from patterned outcomes and not directly observed (Erickson *et al.*, 2015). This paradigm also adequately accommodates a mixed-method approach, hence treating quantitative and qualitative techniques as complementary instruments that can be used to triangulate evidence on the same underlying mechanism (McEvoy and Richards, 2006; Zachariadis, Scott and Barrett, 2013). It also endorses retroductive reasoning aligning with study aim of explaining why CCS came to become the dominant decarbonisation strategy in UK cement industry.

3.3 Research Design

The study employed a convergent parallel mixed-methods design (Creswell and Plano Clark, 2018) nested within a single case study (Yin, 2018). The case is the Peak Cluster – Morecambe Net-Zero CCS network, prompting three embedded analyses which correspond to three research objectives which are, the policy and regulatory framework, the techno-economic dimension and the discursive contestation surrounding the project pipeline. A convergent parallel logic in this instance is preferred because, while the three objectives are conceptually distinct, they are empirically interdependent. This is also preferred because while a sequential design would impose a causal ordering, the path dependency literature outrightly rejects it (Kotilainen *et al.*, 2019).

A case-study approach was considered appropriate in responding to the “how” and “why” questions to a rather contemporary phenomenon outside the researcher's control, especially when the phenomenon and context are entangled (Yin, 2018). The selection of a single case for this study is justified on these three grounds: it is revelatory (world’s largest cement CCS project), spatially representative (demonstrating UK’s industrial cluster strategy) and unique in its implementation (a 200 km onshore corridor crossing two national parks). In alignment with (Flyvbjerg, 2006), generalisation in this case is analytical rather than statistical.

3.4 Study Area

The network covers Derbyshire, Staffordshire, Cheshire, and the eastern Irish Sea and will anchor four major facilities: Holcim’s Cauldon plant, Tarmac’s Tunstead and Hope works, Breedon’s Hope works, and SigmaRoc’s Lime sites, producing about 40 percent of UK cement and lime (Mineral Products Association, 2025). The 200 km onshore pipeline proposed will transport up to 3 Mt CO_2/yr to a coastal facility in Wirral for injection into Morecambe gas fields operated by Spirit Energy (Peak Cluster, 2026). The industries (plants and capture units), infrastructure (pipeline, offshore storage, AGI) and social actors (residents, local authorities, NGOs, industry, regulators) are the three analytical layers around which the study structures the sampling logic as shown in Figure 1.

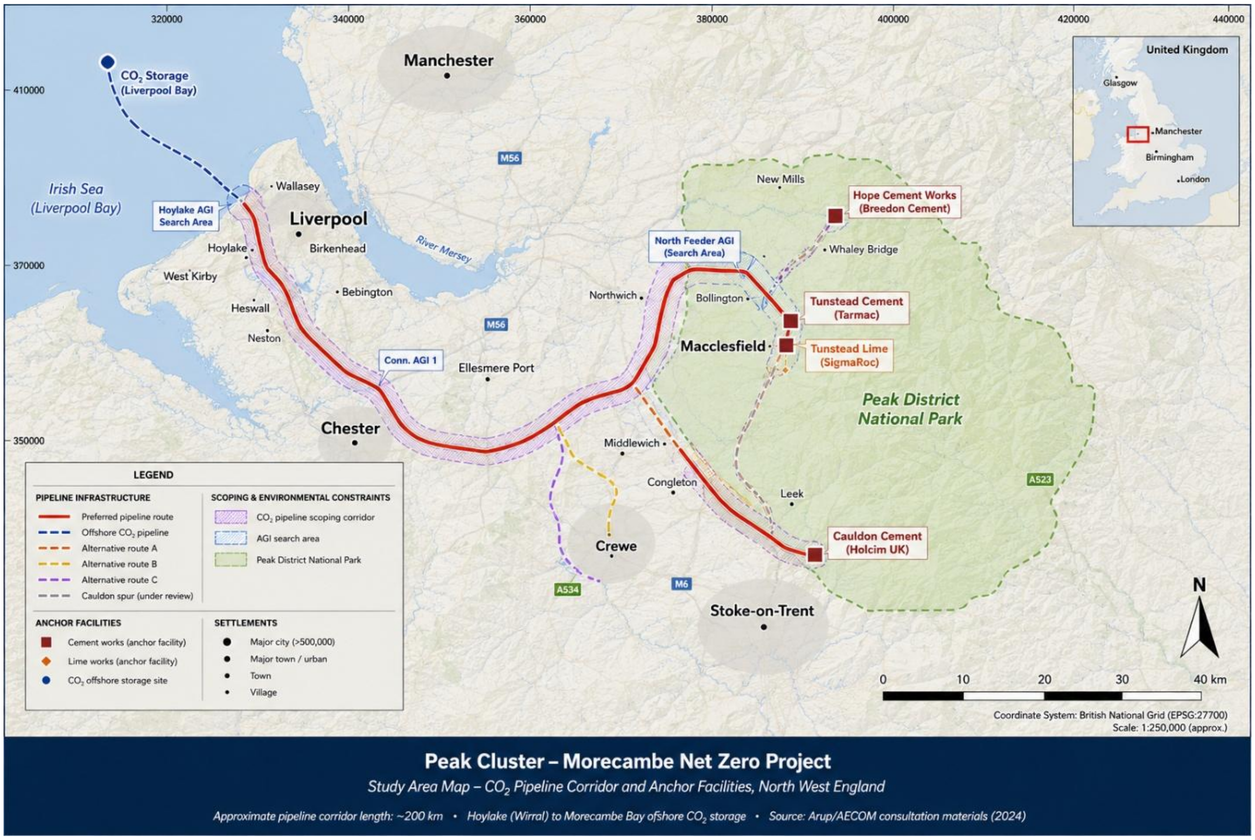

Figure 1 Peak Cluster – Morecambe Net Zero proposed pipeline corridor

Source: Author's construction (Peak Cluster Ltd (2026) consultation materials; Arup/AECOM (2026); Natural Earth 10m vector data (naturalearthdata.com); OS data)

3.5 Policy and Document Analysis

A qualitative document analysis (Bowen, 2009) is used to address research objective 1 in combination with (Bacchi, 2025) "What's the Problem Represented to Be?" (WPR) approach, which helps to uncover the implicit and silent framings that constitute institutional lock-in. Through a purposive criterion-based sampling (Patton, 2015), the document corpus is designed drawing documents from five distinct classes as shown in Table 1.

Fifteen documents are selected as the compact policy corpus and coded P1 – P15, listed in Table 1. Saturation in this study was defined as occurring when additional documents did not yield new narratives or policy mechanisms (Saunders *et al.*, 2018). Documents are imported into NVivo 14. The documents were subject to hybrid coding (Fereday and

Muir-Cochrane, 2006), combined with deductive codes derived from the lock-in typology of (Seto *et al.*, 2016) and inductive codes from the texts. Annotation of each document is as follows: policy instruments, governance, narrative framings, and sequencing logic, with a WPR sub-analysis further interrogating each instrument for presupposed problem, sustaining assumptions and silences.

Institutional path dependency indicators are operationalised utilising the framing of Erickson *et al.,* (2015) as:

i. Temporal density of CCS-favouring instruments relative to alternatives
ii. Public financial commitment per tonne of avoided CO_2
iii. Presence or absence of sunset and pathway-diversification clauses
iv. The rhetorical centrality of CCS in problem representations

Table 1 Policy-corpus document code for analysis

Code	Policy document	Year	Source	Analytical Role
P01	Industrial Decarbonisation Strategy	2021	Business, Energy and Industrial Strategy (BEIS) (BEIS, 2021b)	Roadmap
P02	Net Zero Strategy: Build Back Greener	2021	(BEIS, 2021a)	Roadmap
P03	CCUS Cluster Sequencing Phase-1 Decisions	2021	(BEIS, 2021c)	Sequencing
P04	Industrial Carbon Capture (ICC) Business Model Update (2023)	2023	Department for Energy Security and Net Zero (DESNZ) (DESNZ, 2023)	Contract for Difference (CfD) revenue support (15-yr)
P05	ICC Business Model Update (2025)	2025	DESNZ (DESNZ, 2025)	CfD revenue support; FA forfeiture rule
P06	Powering Up Britain: Net Zero Growth Plan	2022	DESNZ (DESNZ, 2022)	Roadmap
P07	Energy Act 2023	2023	Parliament (Parliament (UK), 2023)	Statutory framework for CCS
P08	Storage of CO_2 (Licensing etc.) Regs 2010	2010	DESNZ (DESNZ, 2010)	Licensing
P09	National Policy Statement EN-1 (rev. 2024)	2024	DESNZ (DESNZ, 2024)	Development Consent Order (DCO) framework
P10	National Policy Statement EN-4 (Gas/Oil Pipelines, CCS)	2024	DESNZ (DESNZ, 2024)	DCO determination framework
P11	MPA UK Concrete & Cement Industry Net-Zero Roadmap	2025	MPA (Mineral Products Association, 2025)	Sectoral roadmap
P12	GCCA Concrete Future: Roadmap to Net Zero 2050	2023	GCCA (GCCA, 2023)	Sectoral roadmap
P13	CCC Progress in Reducing Emissions (Annual Report)	2025	Climate Change Committee (Climate Change Committee, 2025)	Advisory
P14	Peak Cluster Phase-1 Consultation Materials	2026	Peak Cluster (Peak Cluster, 2026)	Pre-DCO consultation
P15	NWF Investment Decision: Peak Cluster	2025	National Wealth Fund (National Wealth Fund, 2025a)	Equity (cornerstone)

3.6 Techno-Economic and Opportunity Cost Analysis

A comparative marginal abatement cost (MAC) benchmarked against alternative decarbonization pathways is used to address objective 2. The MAC framework is used as an instrument to rank mitigation options by cost per tonne of CO_2 avoided (Hasanbeigi *et al.*, 2013; Strunge *et al.*, 2022a; Glenk, Meier and Reichelstein, 2025). Its limitations include independence assumptions, neglect of interactions and blindness to distributional effects (Kesicki and Ekins, 2012), which are addressed in the study through portfolio interaction effects and a regret region extension.

Data for analysis was drawn from peer-reviewed literature, IEA and IEAGHG technical reports, GCCA 2050 datasets, the MPA roadmap, UK Emission Trading Scheme (ETS) plant-level disclosures, and operator sustainability reports. Where primary data are unavailable, a comparable parameter range from similar techno-economic assessments is utilised (Gardarsdottir *et al.*, 2019; Voldsund *et al.*, 2019; Strunge *et al.*, 2022b; Sherif *et al.*, 2025) and triangulated across three or more sources. Financial values used in the analysis are expressed in 2025-2026 USD, converted from sterling at 1.32 USD/GBP.

Table 2 Variables and operational definitions

Variable	Symbol	Operational Definition	Unit
Process emission	E_p	CO_2 emitted from lime calcination	kg CO_2/t clinker
Thermal intensity	T_i	Specific energy at clinker temperature	GJ/t clinker
Full chain CCS cost	C_{ccs}	Levelised capture + transport + storage cost	USD/t CO_2
Subsidy variable	S_g	Per-tonne public subsidy	USD/t CO_2
UK ETS Price	P_c	CO_2 allowance price across 3 scenarios	USD/t CO_2
Energy Penalty	η_p	Extra energy per tonne CO_2 captured	GJ/t CO_2

The levelised cost of CO_2 avoided is expressed mathematically as:

$$LCO_{2ACCS} = (C_{cap} + C_{trans} + C_{stor} + \eta_p * EF_e) - S_g$$

Where EF_e*:* projected grid emission factor

The breakeven carbon price (P_c) is solved for by setting LCO_{2ACCS} equal to carbon price (P_c) when subsidy (S_g) = 0. Sensitivity analysis combines one-at-a-time perturbations with a 10,000 iteration Monte Carlo simulation over empirical parameter ranges aligning with the work of Strunge *et al.,* (2022).

CCS technology is benchmarked against four alternative pathways drawn from chapter 2;

i. Clinker substitution via LC3 and other SCMs (Scrivener, John and Gartner, 2018; Sherif *et al.*, 2025)
ii. Thermal-fuel switching to biomass, waste fuels and hydrogen
iii. Electrification, where commercially demonstrated
iv. Novel low carbon binders (Brimstone, Sublime, TerraCO2).

These pathways are evaluated against abatement potential, levelized cost, readiness of the technology and opportunity cost index defined as cost differential to CCS weighted by abatement scale. The results are visualised as a comparative MAC curve and a portfolio cost frontier, permitting identification of the regret region, hence scenarios within which CCS deployment forecloses cheaper aggregate decarbonization options

3.7 SLO and Social Media Discourse Analysis

The first component of objective 3 focused on a digital discourse analysis of content opposing the Peak Cluster project. Facebook is chosen because preliminary scoping identified active opposition groups and parish-affiliated pages on the platform. After all, it is a dominant medium for place-based community organising in the network's rural localities (Bossetta, 2018). This strand draws on three approaches. Qualitative content analysis (Schreiber, 2013), critical discourse analysis primarily adapted for social media (Fairclough, 2013; Khosravinik, 2017) to examine how ideology shape online communication, and elements of digital ethnography (Pink *et al.*, 2016) providing the tools for stakeholder engagement in virtual "town squares".

3.7.1 Sampling and Corpus Construction

Purposive multistage sampling was employed. The first stage involved the identification of eligible public pages via Boolean search combinations of project terms (“Peak Cluster”, “CO_2 pipeline”, “Morecambe Net Zero”) to build the corpus. Pages must be publicly accessible and contain at least one post referencing the project, and show substantive engagement. Closed groups are excluded on ethical grounds (Townsend and Wallace, 2016), and for any closed group where information was sourced, it was preceded by a written consent. The second stage involved extracting posts and top-level comments within the window from January 2023 to the data cut-off in May, 2026. Stage three applied a relevance filter, and adequacy of sample was judged by information redundancy (Malterud, Siersma and Guassora, 2016), closing the corpus when 3 successive coding cycles produce no new themes.

3.7.2 Coding and Analytical Procedure

Manual data collection with a structured template capturing (post ID, page type, date, text, media type, engagement metrics, comment text). Data storage was in a password-protected NVivo project, held in a University-administered, encrypted storage infrastructure. A thematic reflexive analysis in six phases (Braun and Clarke, 2019) entailing familiarization, initial coding, theme generation, review, definition and naming which was supported by a Faircloughian discourse layer was applied to the finalized theme (text, discursive, practice, social practice). These themes were organized using SLO dimensions (legitimacy, credibility, trust) (Thomson and Boutilier, 2011) and risk perception axes (Wallquist, Visschers and Siegrist, 2010). Findings are operationalized using three indications: frame indicators (dominant frames such as procedural injustice, ecological risk, distrust of operators), sentiment indicators (polarity and intensity), and network indicators (cross-page intertextual references).

The framework method (Gale *et al.*, 2013) is used for the analysis producing a structured matrix permitting cross-case comparison. A continuum of four distinguishing ascending levels (Thomson and Boutilier, 2011): withholding, acceptance, approval and psychological identification, is built operationalizing SLO with stakeholder accounts positioned on this continuum.

3.8 Mixed Methods Integration

The integration employed three mechanisms. *Connecting*:social media corpus. *Merging:* MAC results, policy analysis matrix and social media themes and SLO continuum arrayed in a single joint display matrix for inspection of convergence, complementarity and dissonance (Guetterman, Fetters and Creswell, 2015a). *Building:* meta-matrix interrogated to best account for observed patterns consistent with the critical realist approach in Section 3.2.

3.9 Ethical Considerations

The study follows the Economic and Social Research Council (ESRC) Framework for Research Ethics (UKRI, 2024) and the Association of internet Researchers guidelines (Franzke *et al.*, 2020). For interviews, written informed consent is obtained and anonymity is preserved. While social media data is public, contextual integrity is preserved. Data is stored on the University infrastructure and deleted per institutional policy.

3.10 Trustworthiness and Validity

A four-criteria approach was adopted to evaluate qualitative strands, as shown in Table 3 in line with (Lincoln and Guba, 1988) approach, while quantitative uncertainty is addressed using Monte Carlo sensitivity coupled with transparent data sources and ranges.

Table 3 Lincoln and Gaba's (1985) trustworthiness criteria operationalized

Criterion	**Operational strategy**
Credibility	Triangulation across documents, social media, and interviews. Checking with willing interviewees
Transferability	Detailed description of context, stakeholder, policy environment, explicit articulation of scope conditions
Dependability	Methodological audit trail; version-controlled codebook, documented saturation decisions
Confirmability	Reflexive journaling, transparent positionality, and independent double-coding

Triangulation operates at three levels: data across documents, social media discourse, and techno-economic modelling, and finally, theory, which simultaneously applies path dependency and the SLO framework.

3.11 Limitations

The study has four limitations worth acknowledging. Firstly, the project is targeted no earlier than 2028 (National Wealth Fund, 2025b), so this study only captures the consenting phase dynamics. Secondly, there is substantial uncertainty in CCS techno-economic, though mitigated but not eliminated by Monte Carlo Sensitivity. Thirdly, the Facebook corpus overrepresented the loud networked voices. Finally, researchers' commitment to lock-in and SLO inform the design, while flexible journaling and peer debriefing disciplined these commitments.

CHAPTER FOUR

4.0 RESULTS AND DISCUSSION

This chapter presents empirical findings of the study with its interpretation through the conceptual lenses developed in Chapter 2 with analyses around the three research objectives. Section 4.2 discusses the findings of the qualitative document analysis, and Section 4.3 presents the findings from the techno-economic and opportunity cost analyses, while Section 4.4 presents the pilot public discourse and SLO analysis.

4.1 UK CCS Policy and Institutional Path Dependency

4.1.1 Indicator-Level Findings

Drawing from the lock-in indicators defined in Section 3.5, the dominance of red (strong) in Figure 2, especially in the right two columns, is the main empirical finding. Across the policy corpus, CCS framing was prevalent, with 100% of documents code indicating “High” or “Total” on CCS centrality. Alternative decarbonisation pathways is conspicuously thin, with 60% of documents either marginally referencing alternative pathways to decarbonisation or omitting it entirely. Pathway flexibility coded as “Sunset clauses” which Foxon (2013) and Rietig and Laing (2017) considered critical is represented in 13.3% of the documents, and mentioned partially at best.

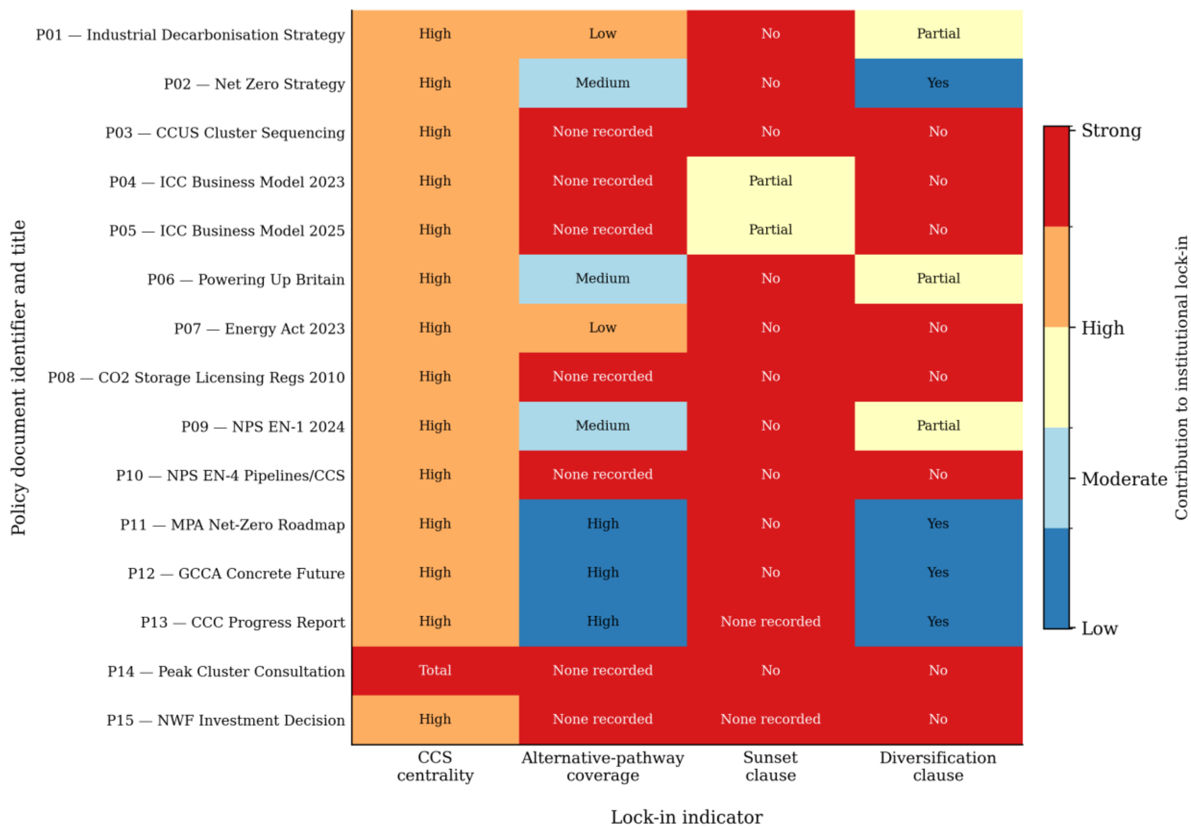

Figure 2 Policy-document lock-in heatmap for P01-P15

Source: author coding of policy corpus, NVivo 14.

The clustering of red across the document is the most important pattern, which is not anomalous to a single document but is dominant across different instrument classes. Peak Cluster (P14) documents display the strongest lock-in profile. P07 – P10, denoting statutory and planning instruments, show similar patterns, showing high centrality on CCS deployment and weak or absent sunset clauses or diversification mechanisms. The least locked-in, comparatively, was industry roadmaps (P11 – 12) retaining portfolio logic, even though they positioned CCS as a residual-emissions backstop. This overall pattern supports a path-dependency claim because lock-in is not concentrated in a single document but distributed across the policy system.

The composite path-dependency composite score (*cs*) on the 0-7 scale defined in Section 3.5 is reported in Figure 3. The corpus mean is 4.40, with P14, the Peak Cluster Phase-

1 consultation materials with the maximum score. This outcome is theoretically unsurprising as project consultation materials presuppose the necessity of the project. More striking is the cluster of strategic and statutory documents, such as the CCUS cluster sequencing (P03), ICC Business Model (P04 – P05), Storage of CO_2 regulations 2010 (P08), and the EN-4 National Policy Statement on Gas/Oil Pipelines and CCS (P10). These strategic statutory documents all scored above the corpus mean and exhibited no diversification clauses and made no sunset provisions.

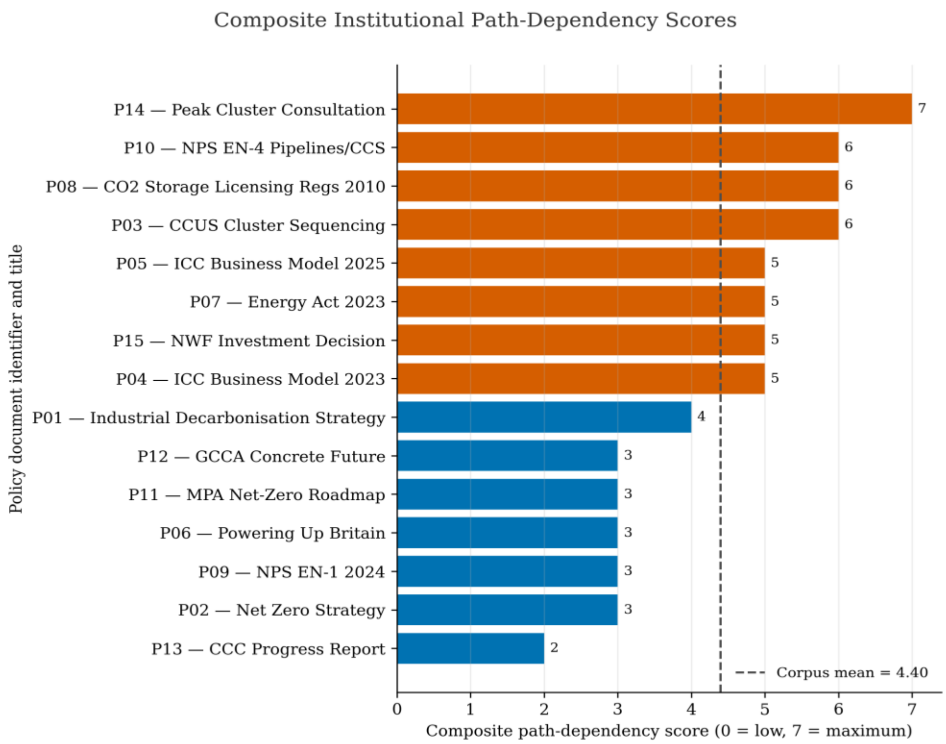

Figure 3 Composite Institutional Path Dependency Score (P01 – P15)

The ranking clarifies by indicating where institutional lock-in accumulates. P14 scores the maximum (*cs = 7*) because the consultation frame places project necessity outside the scope of discourse. P03, P08 and P10 score highly (*cs = 6*) due to a different reason. They encode sequencing, licensing and planning presumptions less visible in the public yet durable institutionally. This is also indicative of the fact that institutional lock-in is strongest where legal authority and public finance (P04, P05, P15) are most directly engaged.

4.1.2 Representation of Problem and Silences

Through the WPR analysis, it is understood that there is a stable representation of the problem across policy documents. Decarbonising hard-to-abate sectors is framed repeatedly as a problem of investor uncertainty for a known technology, rather than a problem of selection of one technology amongst competing pathways. The ICC Business Model Update (DESNZ, 2025) document demonstrates this vividly. The CO_2 strike-price contract and free-allowance forfeiture rule are designed to bridge the gap between CCS cost and the UK ETS reference price yet no similar contractual routes for clinker substitution, supplementary cementitious materials, novel binders, or efficiency-led pathways at scale. The corpus problematizes the financing gap CCS faces, yet leaves unproblematized the question of whether CCS is the lowest cost pathway for decarbonisation of the cement industry.

The silence is rather constitutive and not incidental. The Industrial Decarbonisation Strategy document marginalised alternative binders, the Net Zero Strategy and ICC instruments omit equivalent revenue support for SCM and efficiency, cluster decarbonisation fixes the geographical challenge of decarbonisation finance with no procedures and mechanisms to reopen it if alternatives mature. The National Policy Statement on Gas/Oil and CCS treats CO_2 pipelines as analogous to gas pipelines importing fossil infrastructure, which presumes consent in a rather materially different domain. The National Wealth Fund investment record contains no pathway-diversification condition. Collectively, these omissions develop into an institutional boundary for which CCS is not only a supported option but rather, an option for which the state apparatus has been purpose-built.

4.1.3 Discussing Institutional Lock-In as Self-Reinforcing

Within the context of the UK, these findings substantiate the theoretical claims made by (Erickson *et al.*, 2015; Levidow and Papaioannou, 2016; Seto *et al.*, 2016; Buschmann and Oels, 2019) regarding carbon lock-in. Lock-in emerges as a cumulative product of statute, licensing, planning presumption, business model design, cluster sequencing and public capital; hence, no single document can produce the lock-in effect. Each policy instrument is defensible in isolation, yet collectively, they create a techno-institutional complex, within which CCS is easier to finance, consent and justify than competing decarbonisation pathways.

The implication of this lock-in has a cascading effect. The infrastructure raises the institutional cost of switching pathways. ICC contracts have longer time horizons (15 years), National Policy Statements have embedded planning presumptions, cluster sequencing geographically fixes the allocation of public attention and revenue support. However, if cost of LC3, SCMs and alternative binders falls rapidly over the next decade, which is plausible as the literature reviewed suggests, policy infrastructure will not respond elastically. Institutional commitments are longer-lived than the technological commitment it underwrites. That is clearly the signature of path dependency. The Peak Cluster project should therefore be viewed not as the origin of UK CCS lock-in but as a clear empirical manifestation.

4.2 Evaluating Techno-Economic Viability and Opportunity Cost

4.2.1 Empirical Baseline of Peak Cluster

The annual CO_2 throughput of four of Peak Cluster's anchor facilities was analysed to establish a baseline, as shown in Figure 4. Data were sourced from the UK Pollutant Release and Transfer Register (UK-PRTR) for the 2023 reporting year, covering Scope 1 and 2 emissions within the decarbonisation project's frameworks. From the data, Hope Cement Works contributes 0.985 Mt/yr (43.8%), Tunstead contributes 0.55 Mt/yr (24.6%), Cauldon contributes 0.37 Mt/yr (16.5%), and Buxton Lime contributes 0.34 Mt/yr (15.1%). These emissions aggregate to about 2.25 Mt/yr of CO_2. With a Peak Cluster project-disclosed design throughput of 3 Mt of CO_2 (National Wealth Fund, 2025a; Peak Cluster, 2026), the targeted emissions-reduction goal is consistent with the cluster's aggregate emissions. Subsidy calculations used the 3 Mt/yr disclosed design throughput to maintain consistency in estimates.

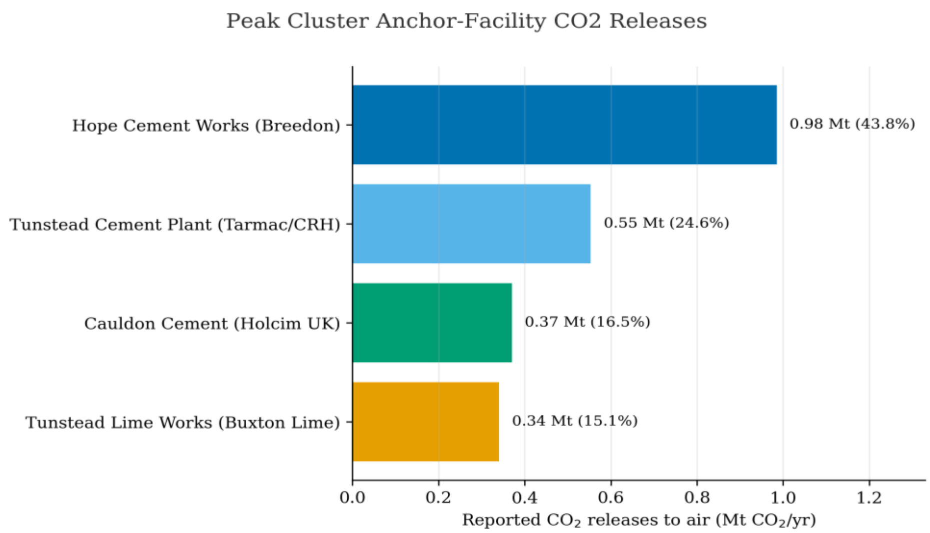

Figure 4: Indicative annual CO2 release for Peak Cluster anchor facilities.

Source: Author construction (UK-PRTR dataset, 2024 released for 2023)

4.2.2 Monte Carlo Analysis of Full-Chain LCO_2A

The full-chain levelized cost of CO_2 avoided (LCO_2A) was computed using the formula specified in Section 3.6, and the results were propagated on a Monte Carlo simulation of 10,000 iterations. Parameter values were drawn from a triangular distribution over empirical ranges in techno-economic literature (Zero Emissions Platform, 2011; Gardarsdottir *et al.*, 2019; Voldsund *et al.*, 2019; Strunge *et al.*, 2022b; IEA, 2023). An evaluation of five capture technologies is represented in the summary statistics Table 4 while Figure 5 represents the full distribution.

Table 4 Monte Carlo Summary statistics for LCO2A, USD/t CO2, no subsidy

Capture technology	**5th Percentile**	**25th Percentile**	**Median**	**Mean**	**75th Percentile**	**95th Percentile**	**Standard Deviation**
Post-combustion Monoethanolamine (MEA)	113.8	123.8	131.4	131.5	139.1	149.7	10.8
Oxyfuel	70.7	77.9	83.2	83.2	88.6	95.9	7.6
Chilled ammonia	96.9	104.7	110.5	110.5	116.1	124.1	8.1
Calcium looping	89.6	97.1	102.8	102.8	108.6	116.3	8.1
Membrane-assisted liquefaction	118.5	126.2	132.2	132.2	138.0	146.1	8.4

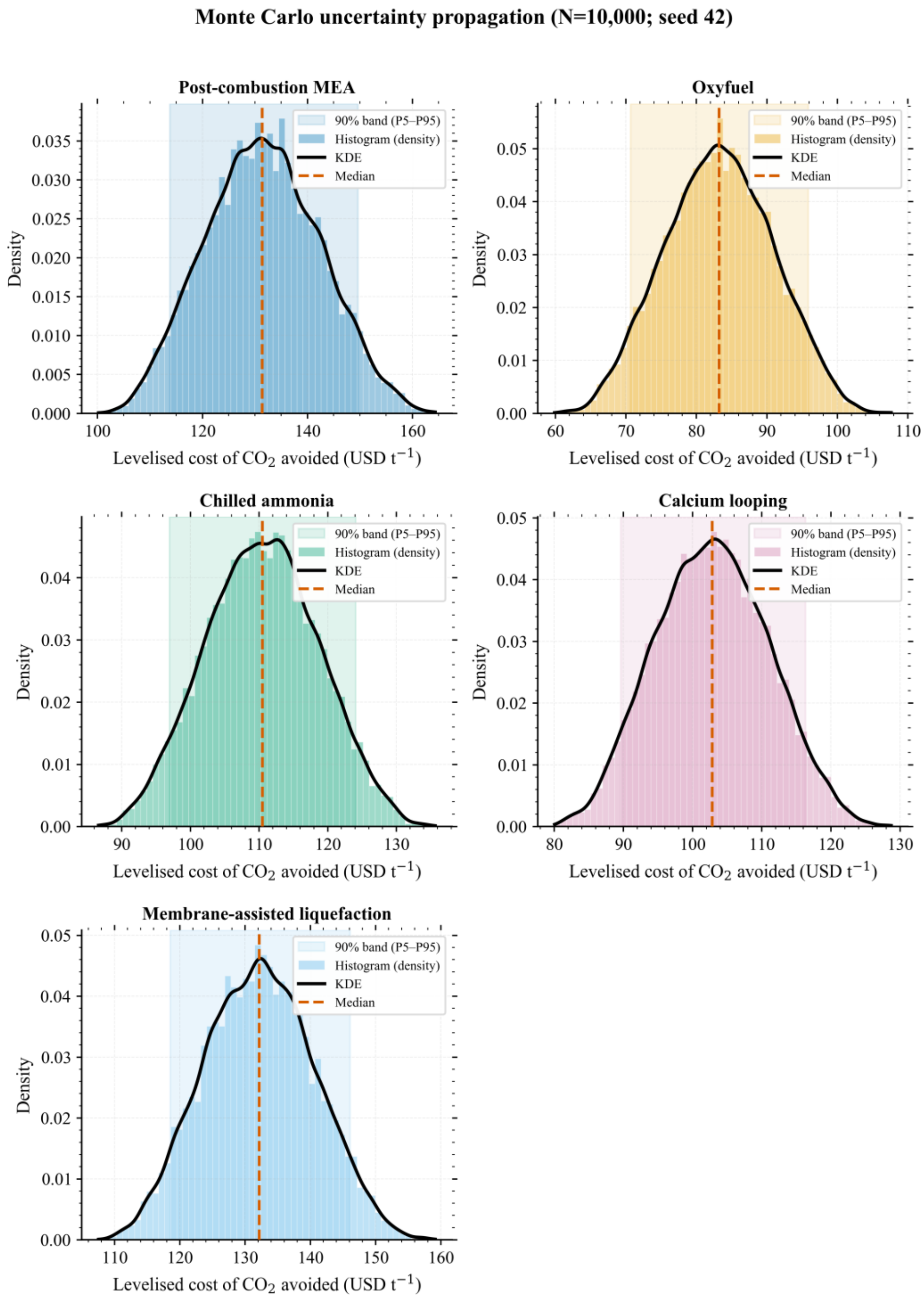

Figure 5 Monte Carlo distribution of LCO_2 by capture technology.

Source: Author's calculation

From Figure 5, it is worth noting that the results do not only present point estimates but also the distributional structure of cost uncertainty. The lowest-cost regions are occupied by Oxyfuel and Calcium looping. However, their uncertainty bands still sit above current UK ETS price on the full-chain cost is included. In contrast, MEA and membrane-assisted liquefaction constitute the upper cost cluster with P5-P95 bands overlapping substantially, implying a subsidy requirement in a low ETS scenario. The inference further strengthens the second objective by clarifying that CCS viability is not only confirmed by technical feasibility but rather a combined interaction of capture technology, cost of carbon price trajectory, and contract design.

This observed pattern in cost uncertainty is not ambiguous. The most mature and plausible retrofit technology for Peak Cluster is Post-combustion MEA, which has a median full-chain LCO_2A of USD 131.4/t with a 90% confidence interval ranging between USD 113.8-149.7/t. A lower median of USD 83.2/t is recorded by Oxyfuel. However, this capture option remains constrained due to lower technology readiness and the absence of large-scale retrofit demonstrations in the cement industry. Overall, the 90% confidence interval for all the capture technologies is above the 2026 UK ETS civil-penalty reference price of USD 65.2/t (GOV.UK, 2025), which makes even the most favourable estimate (oxyfuel) expensive (USD 83.2/t) in comparison.

This finding ties to the economics of CCS to the policy infrastructure analysed previously in Section 2. The Industrial Carbon Capture (ICC) Business Model is a Contract for Difference (CfD) instrument designed for the public and industries/developers in this partnership to settle the difference between the strike price and the UK ETS reference price. At a median cost of USD 131.4/t for MEA and a UK ETS price of USD 65.2/t, an implied subsidy of USD 66.2/t is expected. For a 3 $MtCO_2$/yr project, this runs into USD 198.5 million per year or USD 2.98 billion in 15 years on an undiscounted basis. With oxyfuel, the annual cost saving is approximately USD 4.1 million, but it should be contingent on the ability to be deployed at scale, given the lower readiness. Modelling the subsidy requirement for the viability of the project is therefore not out of the ordinary, but rather a need to understand the gap between the set carbon price by the UK and the chosen path, as well as the policy architecture in place to write off this difference.

The volatility of the UK ETS further reinforces the need for this analysis as shown in Figure 6. ETS prices for CO_2 can rise as high as USD 116.12/t (2022) or fall as low as USD 32 (2022). Considering a central scenario of USD 118.8/t at a very high carbon price path, MEA subsidy could be USD 12.6/t, or approximately USD 18.8/t annually. CCS viability given these scenarios is therefore contingent on two key concepts: capture cost reduction and a strengthened carbon-price mechanism.

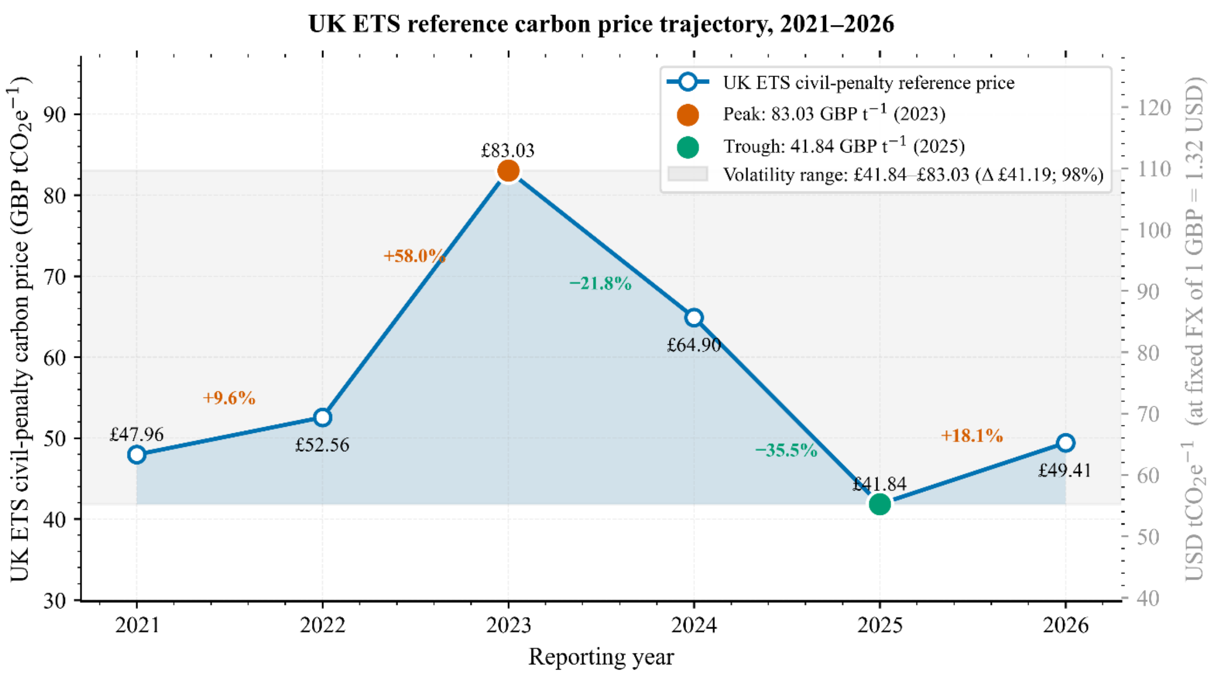

Figure 6 UK ETS reference carbon price trajectory 2021-2026

Source: (GOV.UK, 2025)

4.2.3 Sensitivity Analysis

A tornado sensitivity analysis was conducted for post-combustion MEA technology, considering a base case of LCO_2A of USD 131.7/t. It was found that capture cost dominates the uncertainty structure. Perturbation of ±25% produced a total LCO_2A range of about USD 55.6/t. Contrasting this result was transport contributing USD 4.5/t, storage USD 5.75/t and grid emission factor and energy penalty contributing about USD 5.93/t as shown in Figure 7. This finding is consistent with literature on techno-economic repeated findings that capture, rather than transport or storage, is the dominant cost component in cement CCS deployment (Gardarsdottir *et al.*, 2019; Voldsund *et al.*, 2019).

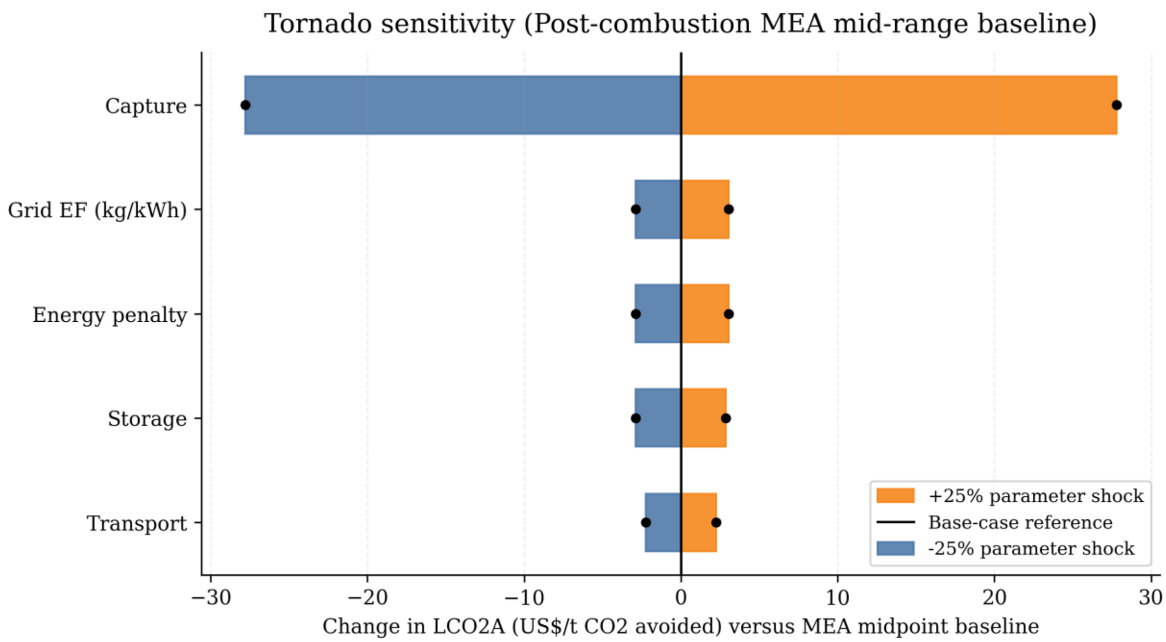

Figure 7 Tornado Sensitivity Shock Analysis for post-combustion MEA

These findings have a wider implication for policy. The most dominant uncertainty (capture cost) indicates that pipeline optimisation and storage economics cannot individually close the subsidy gap. There is still an ongoing debate on whether learning by doing can reduce the capture cost quickly enough to justify long-lived public contracts. Learning rates in cement CO_2 capture for now remain low (Faber *et al.*, 2022), which rests the subsidy case on the technological learning assumption, which is not explicit in policy documents.

4.2.4 Comparative Marginal Abatement Cost Curve (MAC)

The comparative abatement potential presented in Figure 8 represents the different cement decarbonisation pathways as indicated in Section 3.6. The pathways are ordered by median cost and represented from left to right, with bar widths proportional to abatement potential. This information is drawn from GCCA's 2050 roadmap (GCCA, 2023) and MAP decarbonisation roadmap (Mineral Products Association, 2025). The UK ETS price scenarios (UK ETS 2026 civil penalty = 65.2/t, Climate Change Committee

(CCC) recommendation = USD 118.8/t, Net Zero-consistent shadow price = USD 184.8/t) are represented by three horizontal reference lines (Climate Change Committee, 2025; GOV.UK, 2025). In the UK cement and concrete industry context, the first 7.7 $MtCO_2$/yr abatement potential delivered by the different pathway median cost ranges is as follows: efficiency and best practice (1.5 $MtCO_2$/yr at USD 6.6/t), clinker substitution through LC3 and SCMs (3.2 $MtCO_2$/yr/ at USD 33.0/t), alternative fuels (1.8 $MtCO_2$/yr at USD 37.0/t), biomass co-firing (1.2 $MtCO_2$/yr at USD 72.6/t). This pathway is well within the UK ETS price range. CCS oxyfuel abatement enters at a median cost of (3 $MtCO_2$/yr at USD 83.2.t) and novel binders at (0.8 $MtCO_2$/yr at USD 112.2/t). CCS post-combustion MEA has a median cost of (3 $MtCO_2$/yr at 131.4/t). Electrification and hydrogen kiln options occupy the upper curve at USD 158.4/t and USD 171.6/t respectively.

Two areas of concern arise from this outlook. Firstly, if the ETS price stabilises at the 2026 level (USD 65.2), four pathways on the left become competitive without subsidy and a CCS lock-in will retain a subsidy gap of USD 20-70/t. Secondly, if ETS price rises to the CCC trajectory (USD 118.8), the lower cost portfolios like efficiency, SCMs and alternative fuels can now abate 8.7 $MtCO_2$/yr (60% of sector potential). This will, in turn, make the marginal value of an additional unit of CCS abatement smaller than the subsidy required to deliver it. This will be an opportunity cost objection to CCS deployment because public finance now gets to compete with far cheaper abatement options per pound of public money.

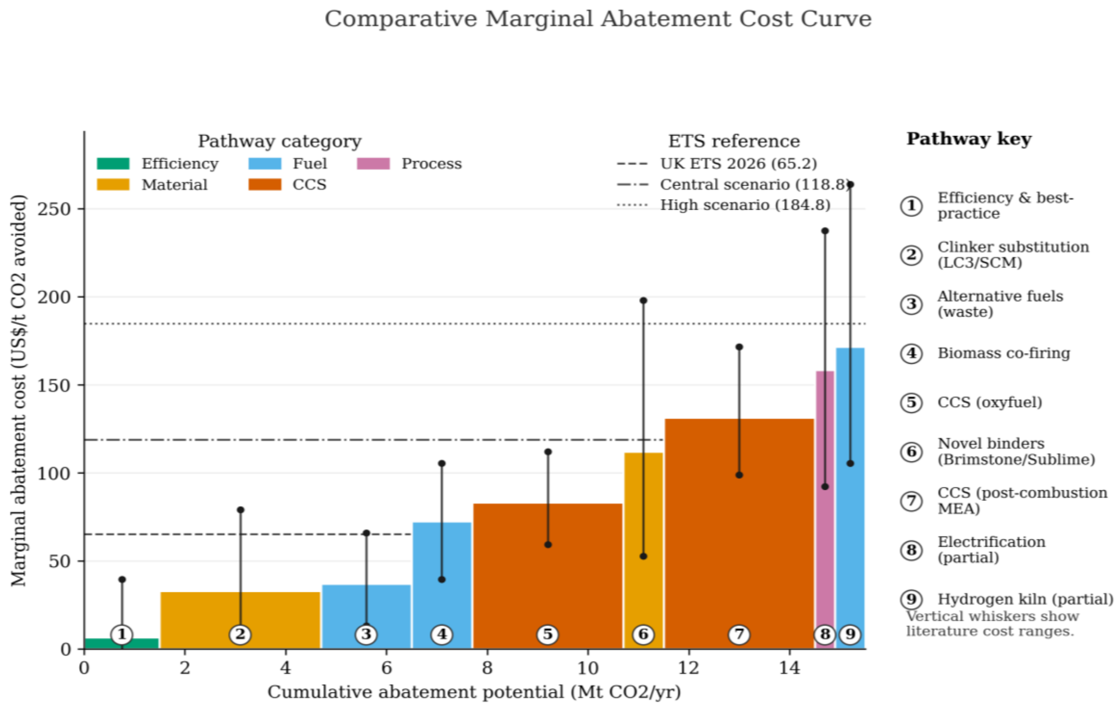

Figure 8 Comparative marginal abatement cost curve, UK cement sector.

Bar height represents the median abatement cost, and error bars show the cost range across techno-economic literature.

4.2.5 Discussing When CCS is an Optimal Investment for the Public

The MAC analysis does not categorically deem CCS as a bad investment. The discourse on CCS investment needs to be analysed through three lenses. First, the abatement potential of the lower cost technologies (efficiency, SCMs and alternative fuels) is bound by the availability of raw materials (calcinated clay, scrap fly ash, biogenic waste streams) and cannot scale indefinitely. The residual process emissions from clinker calcination also remain at 60% of total sector emissions under aggressive substitution scenarios (Sherif *et al.*, 2025). Secondly, the proposed alternatives are equally subject to deployment friction (LC3 supply chain, biomass sustainability constraints), which is a concern not addressed by the MAC. Thirdly, the Peak Cluster's unique selling point is based on securing a UK supply of cement and lime in a context where carbon leakage to imports is an active policy concern, which has a value beyond MAC framing.

The MAC does, however, establish empirical robustness on the question of opportunity cost. Even if we accept CCS's strategic and bounded-substitution role, the public finance responsibility case for the cluster sequencing geography and ICC business model exclusively to CCS, with no parallel revenue support for instruments for SCMs or efficiency measures, is weaker than the policy documents in Section 4.2 suggest. This economic finding points to one conclusion: lock-in is not merely a description of policy instruments, but could also lead to the misallocation of public finance within the broader decarbonisation portfolio.

4.3 Public Perception and Social Licence to Operate

4.3.1 Corpus cleaning and analytical procedure

About 197 online policy items related to the Peak Cluster-Morecambe Net Zero project were collected, spanning January 2023 and 18 May, 2026. After cleaning the data and eliminating duplicates, the corpus yielded 191 unique records. The corpus comprised Facebook posts (n=127, 66%), campaign and commentary text (n=34, 18%), social post (n=28, 15%) and Facebook-referenced news articles[1] (n=2, 1%). The document corpus was coded manually to determine stance, position on the SLO continuum, the frames (primary and secondary), three binary concern indicators (policy, environmental, and economic), and trust posture. A computational VADER sentiment score was generated for triangulation only. The adequacy of the sample was determined by informational redundancy when three successive coding produced no new themes (Malterud, Siersma and Guassora, 2016).

4.3.2 Social license and stance

Figure 9 indicates that the cleaned corpus is overwhelmingly concentrated in the two registers of the SLO continuum. Analysing the 186 non-proponent posts, 91 (48.9%) sit at withholding, 94 (50.5%) at conditional/scrutiny, and 1 (0.5) sits at acceptance. None of the items reached trust-based approval that has been theorised as the upper boundary of SLO (Thomson and Boutilier, 2011). The manual coding also mirrored this picture with 94 (49.2%) posts coded as outright opposition, 76 (39.8%) at neutral/mixed position, 18 (9.4%) at critical scrutiny, and 1 (0.5%) at conditional/engaged, and only 2 (1.0%)

[1] *The corpus composition extended beyond Facebook posts because opposition migrated into dedicated campaign websites.*

expressed proponent support. The corpus thereof indicated no evidence of a trust-based endorsement across the project despite proponent engagement.

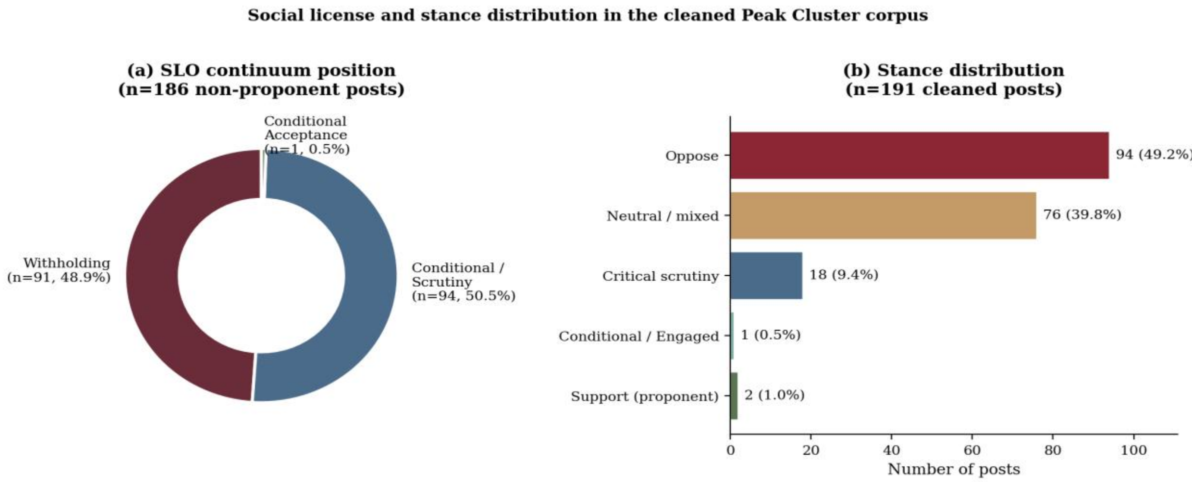

Figure 9 (a) SLO continuum position and (b) manual stance distribution

Source: Public Opinion

A breakdown of the corpus into groups revealed a variance (Figure 10). The most concentrated coalition with a withholding SLO stance was the activists (26/31, 84%). This reflected the organisational role they play in articulating pushback or opposition (Bernstein and Hoffmann, 2018). Elected representatives of various local governments were also concentrated on withholding SLO (27/37, 73%), even though a quarter occupied the conditional/scrutiny position, which is consistent with statutory consultation obligations. The largest bloc of stakeholders, the residents and community voices (n=104) displayed an opposite pattern, with 67% at conditional/scrutiny and 31% at withholding. This indicated that while a majority of residents along the corridor have not yet outrightly rejected the project, the option to withhold endorsement is linked to the procedural reassurance they get, which is consistent with (Stalker *et al.*, 2022) findings that public acceptance of carbon and hydrogen is heavily dependent on controllability, demonstrable monitoring and trust in regulators. Media coverage (n=16) is distributed across all three registers which is found consistent with reportorial framing rather than advocacy.

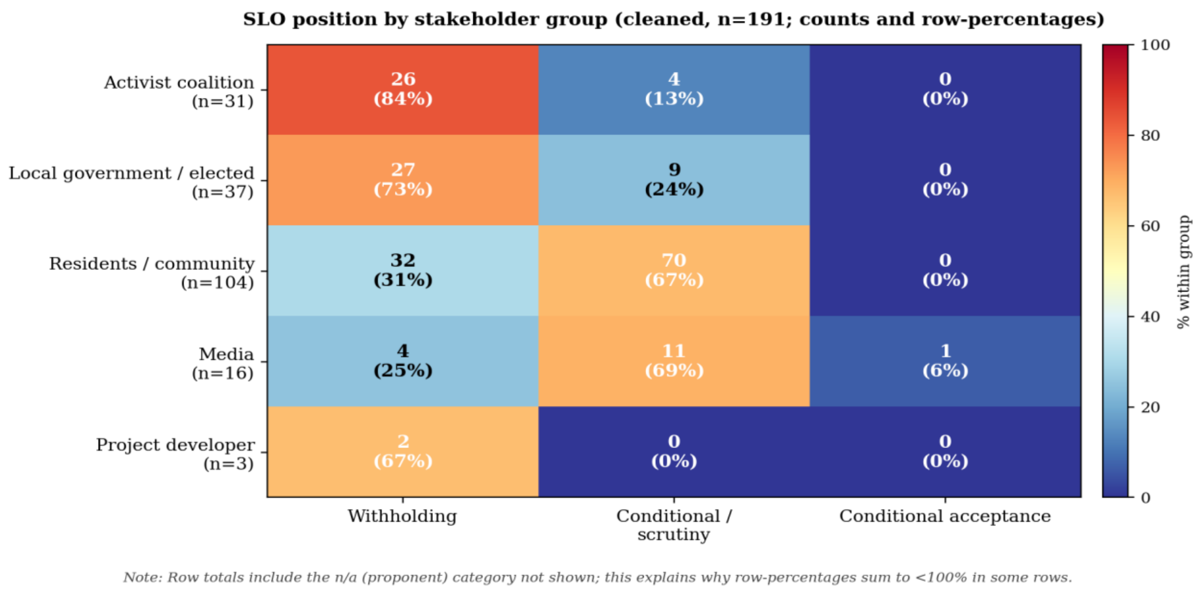

Figure 10 SLO Position by stakeholder group

4.3.3 Three (3) dominant interpretive frames

Employing reflexive thematic analysis (Braun and Clarke, 2019), a three-frame family was identified as representative of about 98% of the corpus (Figure 11 (a)). The modal primary frame was place identity and landscapes (n=93, 49%), followed by procedural and consultation governance (n=65, 34%) and safety and ecological risk (n=29, 15%). Other frames, like Economic, technical legitimacy and trust deficit frames all appear in fewer frames as a primary concern. Economic concerns and trust however appear as a secondary frame. Based on the analysis, the picture painted in public discourse against the Peak Cluster project is one of defence of place and process, not a contest over carbon economics, which aligns with the findings of Lai and Devine-Wright, (2024), that "contested net-zero industrial clusters become arenas in which place-based imaginaries are mobilised against techno-economic ones".

4.3.3.1 Place Identity and landscape (49%)

This place identity positions the Peak Cluster pipeline as an intrusion into landscapes that represent regional identity, such as the Peak District National Park, the Wirral coast, Cheshire farmland and the Staffordshire Moorlands. The lexicon is one of custodianship/stewardship (Devine-Wright, 2009). An activist coalition extract reads as follows:

" Ordinary people … forced to come together to ask the difficult questions about what this project could mean for our countryside, our coastline, our villages, our homes, and our future"

The collective ownership is performed in this frame through repeated use of "our", a possessive pronoun, which presents residents as legitimate guardians pushing back against outside imposition. This extract challenges the first pillar of SLO, legitimacy (Jijelava and Vanclay, 2017), thereby constructing the project as being incompatible with the place. These moments have been described by Buschmann and Oels, (2019) as "discursive turning points" capable of disrupting an otherwise entrenched carbon-locked in trajectory.

4.3.3.2 Procedural and consultation governance (34%)

This frame does not contest the procedural frame of CCS but the manner consent is being sought to advance the Peak Cluster project. It is mostly championed by local government and activist coalition actors, challenging the adequacy of consultation, DCO sequencing, and the perceived asymmetry between project communication resources and community capacity (Witte, 2021). One resident's statement is paraphrased as follows:

"The fact that the Metro Mayor says he will not rush this decision tells you everything; communities were not consulted early enough or meaningfully"

This framing could be consequential in the future because it identifies a remediable deficit. Whereas arguments within the place identity frame cannot be ameliorated/mitigated by better engineering, these procedural concerns at least in principle, can be mitigated by genuine procedural reform (Gough and Mander, 2019).

4.3.3.3 Safety and ecological risk (15%)

This framing is drawn from the unknown risk distinction (Slovic, 1987) to determine that the category of risk associated with a 200km pipeline and offshore Morecambe injection is one for which there is no community track record. The asphyxiant payload of CO_2 to be transported via the pipeline and the permanence of geological storage are unknown components (Wallquist, Visschers and Siegrist, 2010). Several posts also invoked the cancellation of Barendrecht 2010 as a precedent of caution (Dolan, 2020), which was meant to serve two purposes: legitimise opposition by locals and signal project cancellation risks to proponents.

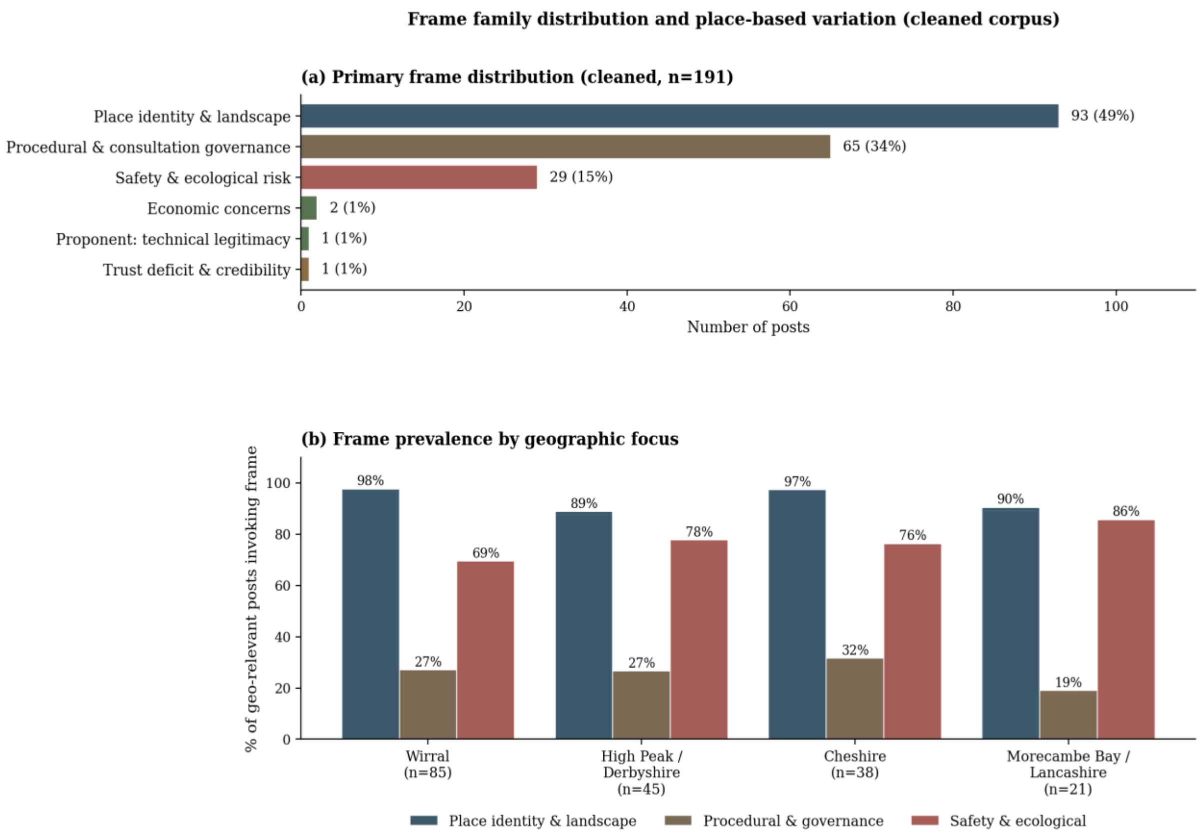

Figure 11 Primary frame distribution of corpus (b) geographic focus of frames[2]

Source: Social media corpus

4.3.4 Geographic variation and temporal clustering

Across the four geographies, three themes dominated as shown in Figure 11 (b). One of the themes, which is near universal (89-98%), is place identity. The safety frame is most concerning and concentrated at Morecambe Bay/Lancashire (86%), Cheshire (76%) and High Peak/Derbyshire (78%), tracking the proximity of these locations to the injection site and onshore pipeline alignment, respectively.

[2] *Post may reference multiple geographical locations*

Comparatively, the procedural framing is weaker in absolute terms but was found consistent across different geographies (19-32%), which suggested procedural concerns generalise across the corridor rather than cluster at any given point (Mabon, Shackley and Bower-Bir, 2014). Wirral, where the Moels Above Ground Installation (AGI) is the only geography where three frames coincide at meaningful prevalence (place 98%, procedural 27%, safety 69%) consistent with the AGI status as it is the most visible point of materialisation along the route.

4.3.5 Concerns dimensions and trust deficit

Through deductive coding, three binary concern indicators and trust postures were produced (Figure 12(a)). Policy concern is almost ubiquitous (148/191, 78%), environmental concern represents two-thirds (129, 68%), with economic concern being the minority (52, 27%). Trust deficit involving questioning the credibility of operators was found in 171 posts (90%), with only 20 (10%) posts actively expressing trust. This aligns with the findings of Gordon, Balta-Ozkan and Nabavi, (2023) towards repurposed energy pipelines where he asserts that public attitude is the principal socio-technical barrier to deployment.

The less prominent status of explicit economic framings is notable when compared side-by-side with the techno-economic findings in Section 4.2. The public discourse, unlike in section 4.2, which centres on cost per tonne abated, breakeven price, and opportunity cost, places emphasis on place, process and precaution. Where the economic frame is missing, it is presented in the form of a subsidy critique as in this extract:

"Public money propping up polluting industries … corporate welfare dressed up as green policy"

This is consistent with the path dependence reading made in Section 4.1 because a significant amount of public discourse describe Peak Cluster as a means to preserve incumbency rather than create transformative pathways, which aligns with the academic critique of policy lock-in (Erickson *et al.*, 2015; Levidow and Papaioannou, 2016; Seto *et al.*, 2016)

The VADER sentiment analysis results (Figure 12 (b)) presents a methodological caution. 139 (79%) posts return a positive computational polarity, yet 94 (49%) of those very posts are coded as opposition in the manual stance. The reason that was found from this analysis is that mobilisation discourse is linguistically positive ("standing together", "protecting what we love", "ordinary people") even when the stance is one of refusal (Khosravinik, 2017). Hence, sentiment polarity in this corpus does not map to stance, implying that manual interpretive coding remains indispensable for SLO research. The VADER results is therefore retained to evidence the mismatch and signal that policy relevant sentiment dashboards built on automated polarity scores could substantially misread the discursive situation around Peak Cluster.

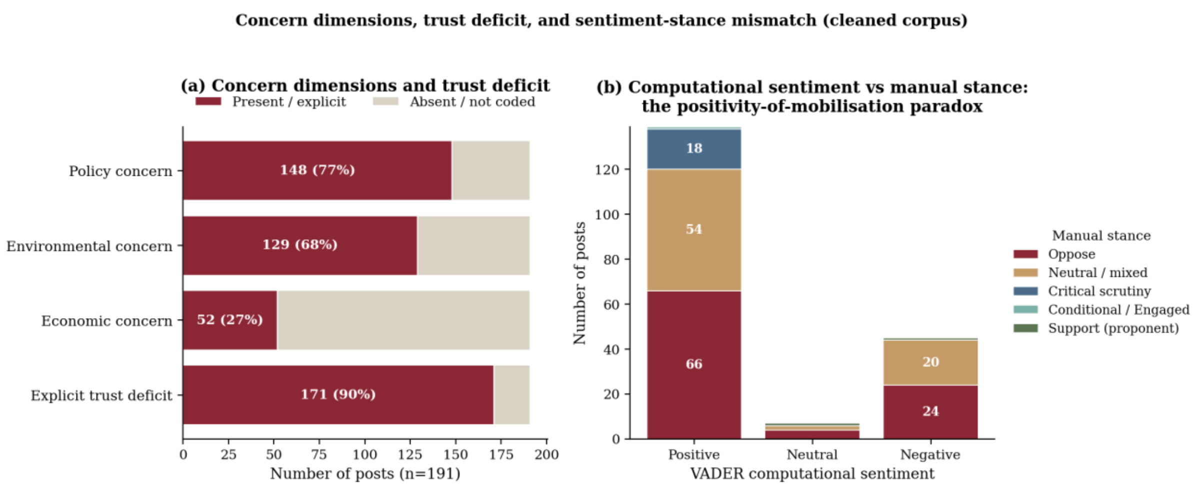

Figure 12 (a) Policy, economic and environmental concerns (b) VADER computational sentiment versus manually coded stance

Source: Social media corpus

4.3.6 Limitations of social media analysis

Firstly, the corpus analysis over-represents the loudest and networked voices while under-representing silent acceptors, thereby characterising this study as a publicly performed contestation rather than the population's sentiments (Townsend and Wallace, 2016). In such networked voices, anybody with a pragmatic view on the project is immediately shot down by the vast majority of the community. Secondly, the Facebook-centric analysis excluded posts from X (formerly Twitter), LinkedIn and offline media, though preliminary scoping identified Facebook as the most dominant place-based community along the corridor (Bossetta, 2018).

4.4 Mixed methods integration

4.4.1 Convergent mixed-methods case study

This approach is grounded in critical realism (Bhaskar, 2008; Fletcher, 2016) and not a stylistic preference but rather an analytical commitment. The objective being sought after is the mechanism that operates simultaneously across institutional, infrastructural and behavioural dimensions, which cannot be directly observed but only inferred by observing patterns as they converge across multiple domains. So, a policy-only study will have described the institutional architecture without showing any contestations, the techno-economic only study will have established the cost gap while treating policy as an external factor and the social media only study explains opposition without contextualising the regime within which it is mobilised. So, the three strands are distinct yet empirically interdependent, hence do not merely supplement but rather jointly constitute the analytical argument.

4.4.2 Operationalising integration of objectives

The three objectives were integrated aligning with (Fetters, Curry and Creswell, 2013) approach on achieving mixed method principles. Linked strands were connected in one joint-display diagram (Guetterman, Fetters and Creswell, 2015b), presented in Figure 13, allowing the inspection of convergence, complementarity and dissonance. Patterns of convergence and dissonance were interrogated to better understand what best accounted for the observed configuration.

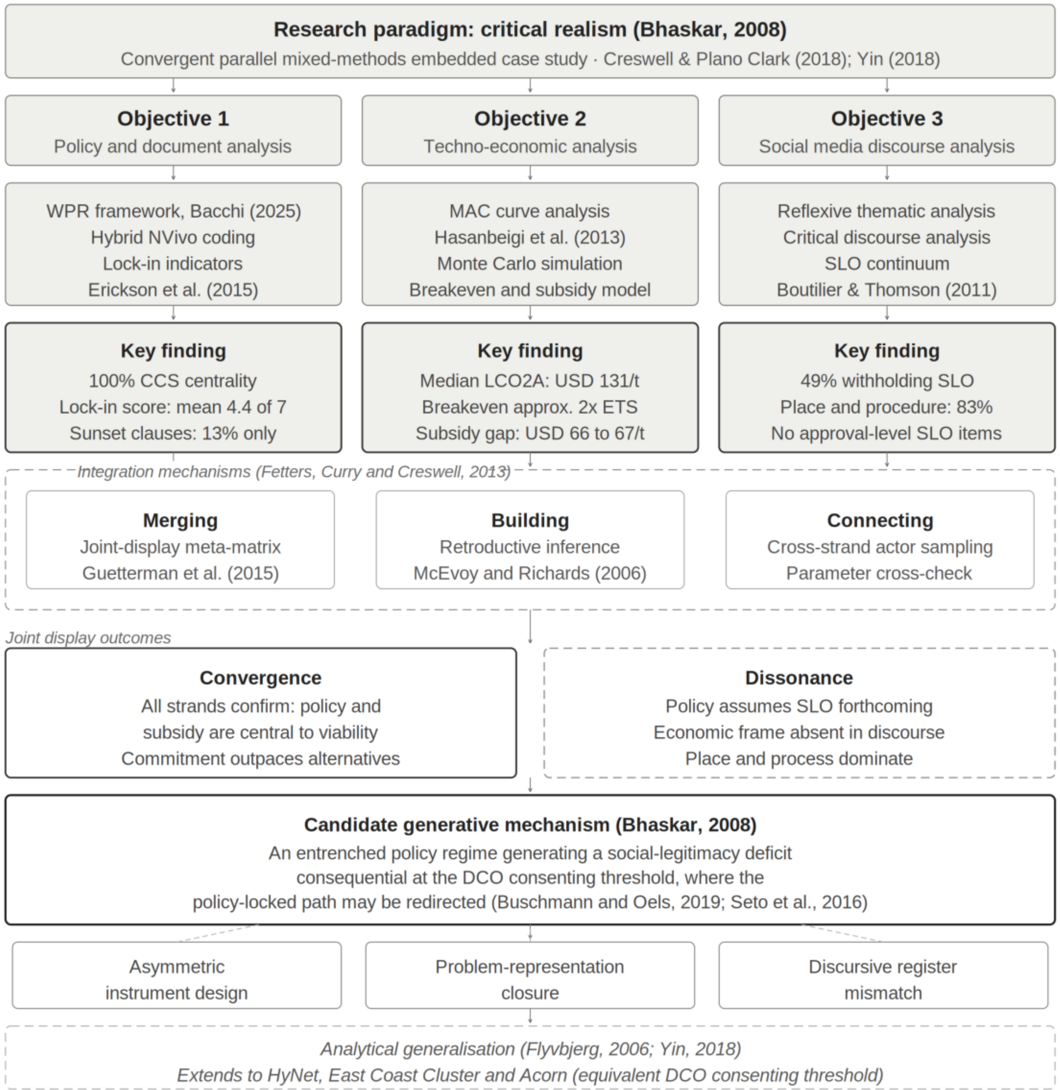

Figure 13 Joint display of convergence, dissonance and generative mechanisms

Source: Author's construction.

4.4.3 What the joint display reveals

One central pattern in Figure 13 is that the three objectives/strands converge on the centrality of policy and subsidy to the project's existence, but also diverge on the legitimacy of resulting commitment. While the policy strand documents an entrenched institutional regime, the techno-economic strand demonstrates that such a regime is crucial to bridge the structural cost gap. The social media strand also reveals that this regime, as interpreted by 90% of stakeholders, is evidence of procedural illegitimacy and not a legitimate response to a technological challenge. This dissonance cannot be treated as noise because the same policy instruments that make Peak Cluster commercially viable are the very same instruments that make it socially illegitimate. However, the institutional regime will not be able to resolve the social license deficit, which means to resolve this, project proponents need to alter that logic.

CHAPTER FIVE

5.0 CONCLUSION AND RECOMMENDATION

5.1 Summary of findings

The thesis sought to evaluate policy drivers, economic viability and public perception of CCS in the UK cement sector using the Peak Cluster-Morecambe Net Zero project as its empirical case. The findings across these three parallel strands revealed a coherent yet troubling picture of a project that is institutionally entrenched, economically contingent on public subsidy, and socially contested in a way the governing framework is yet to acknowledge.

The policy analysis has demonstrated that CCS has a self-reinforcing architecture, which is embedded in the UK industrial decarbonisation pathway. With a mean lock-in score of 4.4/7 across 15 document corpora, a diversification clause was only present in 13% of policy instruments, indicating that institutional infrastructure not only favours CCS, but it also reinforces deliberation needed to justify CCS as a preferred decarbonisation option.

The techno-economic analysis shows that institutional commitment cannot be justified on commercial grounds alone. The median levelised cost of CO2 avoided for post-combustion MEA is approximately USD 131/tonne, compared with a 2026 UK ETS reference price of USD 65/tonne, implying a subsidy of USD 66/tonne and an annual public cost of USD 199 million at the 3 million tonne design throughput by Peak Cluster. In comparison with alternative decarbonisation pathways such as LC3 and SCMs, a meaningful abatement cost, which is substantially below the CCS breakeven without pipeline infrastructure, geological storage, or long-term liability, is attained. The opportunity cost forgone under a CCS-only commitment is estimated between USD 20 and USD 70 per tonne under plausible deployment scenarios under the current UK 2026 ETS carbon price scenario.

The social media discourse analysis revealed an absence of a secured social license at any level above conditional withholding. Out of the 191 unique records, 49% of non-proponent items sit at withholding, 51% at conditional scrutiny, with no items reaching approval or psychological identification. The dominant frames are place identity and landscape (49%), procedural and consultation governance (34%) and safety and ecological risk (15%). The instruments considered precursors for the viability of Peak cluster are the very instruments that community actors consider evidence of procedural pre-determination and corporate welfare.

The integration of these three objectives/strands produced a central theoretical contribution, which is the identification of a social pathbreaker mechanism within the carbon lock-in framework. The Development Consent Order (DCO) consenting is therefore the threshold responsible for institutional pivot and testing of the implicit assumption of the policy regime, serving as the lever where path dependence and path change are simultaneously possible.

5.2 Recommendations

The findings from the study generated six specific recommendations addressed to three actors (Project Proponents, UK Government, consenting authority) with the most leverage over project trajectory.

For the project proponents:

1. Initiate genuine pre-DCO procedural reform immediately because the procedural process represented in the corpus has been deemed inadequate, asymmetric and pre-determined. An independent review of consultation quality should be commissioned and engagement timelines should be extended beyond statutory minima and establish a community liaison mechanism with decision-influencing capacity. This is because 51% currently at conditional scrutiny require procedural credibility, not technical arguments, to move upward on the social license continuum.
2. Commission and publish a comprehensive, independent verification of full CO_2 transport and storage risk envelope. The safety and ecological risk frame is highly concentrated in Wirral and Morecambe Bay discourse since residents are at the closest proximity to the proposed above-ground installation. This should be independent publicly accessible risk assessments and not communication materials produced by project proponent.

3. Affected communities should be engaged using place sensitive methods that acknowledges the territorial identity, not just the infrastructural risk. Risk communication is not enough to address the place identity concerns alone. These place-based social impact assessments should treat landscape character and community identity as major concerns and not residual planning issues.

For the UK Government:

1. Introduction of a pathway diversification provision and sunset review clauses into all future industrial decarbonisation instruments. The absence of such in 87% of policy corpus operationalised an institutional lock-in which calls for a formal review obligation at five-year intervals. This will be instrumental in interrupting self-reinforcing logic without undermining investor confidence.
2. Extend CfD-style revenue support instruments to alternative low-carbon cement pathways, including LC3, SCMs, calcinated clay and novel binders. Configuring the current system to only CCS exclusively does not allow the market to determine an optimal abatement portfolio, which essentially implies that these decarbonisation pathways are determined administratively.

For the consenting authority:

1. There is a need for a structured social acceptance assessment which is capable of distinguishing withholding and conditional SLO continuum. Existing planning frameworks treat opposition as a material consideration to be weighed rather than a dynamic license problem to be addressed. Structuring the SLO assessment into conditional blocs will enable the proponents to examine conditions that can be ameliorated thereby generating a basis for consent.

So, while industrial decarbonisation in the UK might be path-dependent, it is not deterministic. The findings indicate that conditional blocs are real, procedural deficits are remediable and alternative pathways to decarbonisation are available. The success of the Peak Cluster project lies in its ability to bridge the gap between commercial viability, social legitimacy based on decisions of proponents, government, and the consenting authority, which are currently being made. The big question is, will those decisions be made in time, before the DCO threshold renders their consequences irreversible?

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APPENDICES

APPENDIX A – POLICY DOCUMENT LOCK-IN SCORING

Table A1 Policy corpus documents summary and lock-in composite score

ID	Document Title	Year	Issuer	Class	CCS Central.	Alt. Coverage	Sunset	Diversify	Score /7
P01	Industrial Decarbonisation Strategy	2021	BEIS	Strategy	High	Low	No	Partial	4
P02	Net Zero Strategy: Build Back Greener	2021	BEIS	Strategy	High	Medium	No	Yes	3
P03	CCUS Cluster Sequencing Phase-1	2021	BEIS	Programme decision	High	—	No	No	6
P04	ICC Business Model Update (2023)	2023	DESNZ	Business model	High	—	Partial	No	5
P05	ICC Business Model Update (2025)	2025	DESNZ	Business model	High	—	Partial	No	5
P06	Powering Up Britain: Net Zero Growth Plan	2023	DESNZ	Strategy	High	Medium	No	Partial	3
P07	Energy Act 2023	2023	Parliament	Primary legislation	High	Low	No	No	5
P08	CO_2 Storage Licensing Regs 2010	2010	Parliament	Regulation	High	—	No	No	6
P09	NPS EN-1 (rev. 2024)	2024	DESNZ	Planning policy	High	Medium	No	Partial	3
P10	NPS EN-4 (Gas/Oil Pipelines, CCS)	2024	DESNZ	Planning policy	High	—	No	No	6
P11	MPA Net-Zero Roadmap	2025	MPA	Industry roadmap	High	Medium	No	No	3
P12	GCCA Concrete Future 2050	2023	GCCA	Industry roadmap	High	Medium	No	No	3
P13	CCC Annual Progress Report	2025	CCC	Independent advice	Medium	Medium	No	Yes	2
P14	Peak Cluster Consultation Materials	2026	Peak Cluster Ltd	Project documentation	High	—	No	No	7
P15	NWF Investment Decision	2025	NWF	Public investment	High	—	No	No	5

Note: Score /7 = composite path-dependency indicator following Erickson et al. (2015). Mean score across corpus = 4.4. Sunset clause = explicit provision for technology review or instrument termination. Diversify = explicit provision for alternative-pathway revenue support. '—' in Alt. Coverage denotes no mention of alternatives.

APPENDIX B – CLUSTER ANCHOR FACILITY DATA

Table A2 Peak Cluster anchor facility CO_2 emissions 20

Facility	Operator	County	Type	PRTR ID	Permit Ref.	CO_2 (Mt/yr)	Share (%)	Rep. Year
Hope Cement Works	Breedon Cement Limited	Derbyshire	Cement	EW_EA-16879	EPR/BP3731VJ	0.985	43.8%	2023
Tunstead Cement Plant	Tarmac Cement Limited	Derbyshire	Cement	EW_EA-6933	EPR/XP3532DP	0.553	24.6%	2023
Cauldon Cement	Lafarge Cauldon Ltd (Holcim UK)	Staffordshire	Cement	EW_EA-365	EPR/TP3334AW	0.370	16.5%	2023
Tunstead Lime Works	Buxton Lime Limited	Derbyshire	Lime	EW_EA-23705	n/a (lime permit)	0.340	15.1%	2023
Anchor Total	Four sites combined	—	—	—	—	**2.248**	**100%**	2023

Source: Defra UK-PRTR full XML dataset, 2024 release (https://prtr.defra.gov.uk/full-dataset), retrieved 01 May 2026.

APPENDIX C – TECHNO-ECONOMIC ANALYSIS PARAMETERS

Table A3 TEA input parameter ranges used in LCO_2A calculation

Parameter	Unit	Low	High	Source
Post-combustion MEA capture	USD/t CO_2	79.00	125.00	Gardarsdottir et al. (2019); IEAGHG (2018); converted at 1.32 USD/GBP
Oxyfuel capture	USD/t CO_2	42.00	73.00	Voldsund et al. (2019); Gardarsdottir et al. (2019)
Chilled ammonia capture	USD/t CO_2	66.00	99.00	Gardarsdottir et al. (2019)
Calcium looping	USD/t CO_2	59.00	92.00	Gardarsdottir et al. (2019)
Membrane-assisted liquefaction	USD/t CO_2	86.00	119.00	Gardarsdottir et al. (2019)
Onshore pipeline transport (200 km, 3 Mt/yr)	USD/t CO_2	4.60	11.90	ZEP (2011); IEA (2020)
Offshore storage (depleted gas reservoir)	USD/t CO_2	5.30	15.80	ZEP (2011); GCCSI (2021)
Energy penalty (amine, post-combustion)	GJ/t CO_2 captured	2.50	4.50	Voldsund et al. (2019); IEAGHG
Grid emission factor (2030+ projection)	kg CO_2/kWh	0.050	0.120	BEIS Green Book; CCC Sixth Carbon Budget
UK ETS scenario – Low (2026 civil penalty)	USD/t CO_2	65.20	65.20	GOV.UK UK ETS Authority (2025)
UK ETS scenario – Central (CCC trajectory)	USD/t CO_2	106.92	130.68	CCC Sixth Carbon Budget
UK ETS scenario – High (IEA Net-Zero)	USD/t CO_2	171.86	197.74	IEA Net-Zero 2050
GBP–USD exchange rate (2025 annual avg)	USD per GBP	1.319	1.319	exchange-rates.org 2025 annual average

All costs in 2024–25 USD/t CO_2

APPENDIX D – MONTE CARLO SUMMARY RESULTS

Table A4 Monte Carlo LCO$_2$A summary statistics

Capture Technology	P5	P25	Median	Mean	P75	P95	SD	Median (USD/t)	Breakeven ETS (USD/t)	Viable at ETS 65?	Implied Subsidy (USD/t)
Post-combustion MEA	113.8	123.8	131.4	131.5	139.1	149.7	10.8	131.4	131.4	No	66.2
Oxyfuel	70.7	77.9	83.2	83.2	88.6	95.9	7.6	83.2	83.2	No	18.0
Chilled ammonia	96.9	104.7	110.5	110.5	116.1	124.1	8.1	110.5	110.5	No	45.3
Calcium looping	89.6	97.1	102.8	102.8	108.6	116.3	8.1	102.8	102.8	No	37.6
Membrane-assisted liquefaction	118.5	126.2	132.2	132.2	138.0	146.1	8.4	132.2	132.2	No	67.0

APPENDIX E – MARGINAL ABATEMENT COST DATA

Table A5 Marginal abatement cost (MAC) curve data for pathways

Pathway Option	Abatement Potential (Mt/yr)	Cost Low (USD/t)	Cost High (USD/t)	Cost Mid (USD/t)	TRL	Category	Cumulative (Mt/yr)	Opp. Cost vs MEA (USD/t)
Efficiency & best-practice	1.5	−13.2	39.6	6.6	9	Efficiency	1.5	+124.8
Clinker substitution (LC3/SCM)	3.2	6.6	79.2	33.0	8	Material	4.7	+98.4
Alternative fuels (waste)	1.8	13.2	66.0	37.0	9	Fuel	6.5	+94.4
Biomass co-firing	1.2	39.6	105.6	72.6	8	Fuel	7.7	+58.8
CCS (oxyfuel)	3.0	59.4	112.2	83.2	6	CCS	10.7	+48.2
Novel binders (Brimstone/Sublime)	0.8	52.8	198.0	112.2	5	Material	11.5	+19.2
CCS (post-combustion MEA)	**3.0**	**99.0**	**171.6**	**131.4**	**7**	**CCS**	**14.5**	**0.0 (baseline)**
Electrification (partial)	0.4	92.4	237.6	158.4	4	Process	14.9	−27.0
Hydrogen kiln (partial)	0.6	105.6	264.0	171.6	5	Fuel	15.5	−40.2

Opportunity cost = cost differential vs MEA (positive = cheaper than MEA; negative = more expensive).

Sources: Strunge et al. (2022); MPA (2025); GCCA (2023); Sherif et al. (2025); Voldsund et al. (2019).

APPENDIX F – UK ETS CIVIL PENALTY CARBON PRICES 2021 – 2026

Table A6 UK ETS statutory carbon prices, 2021–2026

Year	GBP/t CO_2	USD/t CO_2	YoY Change	GOV.UK Source
2021	47.96	63.31	—	GOV.UK UK ETS civil-penalty determinations 2021
2022	52.56	69.38	+9.59%	GOV.UK UK ETS civil-penalty determinations 2022
2023	83.03	109.60	+57.97%	GOV.UK UK ETS civil-penalty determinations 2023
2024	64.90	85.67	−21.84%	GOV.UK UK ETS civil-penalty determinations 2024
2025	41.84	55.23	−35.53%	GOV.UK UK ETS civil-penalty determinations 2025
2026	49.41	65.22	+18.09%	GOV.UK UK ETS civil-penalty determinations 2026

Source: UK ETS Authority (DESNZ, Scottish Government, Welsh Government, DAERA).

APPENDIX G – ANALYSIS CODE REPOSITORY

All Python analysis scripts underpinning the quantitative and qualitative-quantitative findings are deposited in a public GitHub repository. The repository is designed to permit full reproducibility of all computational results, including the Monte Carlo simulation, the marginal abatement cost curve, and the UK ETS subsidy breakeven calculations.

Repository	https://github.com/Authenticscholar/peak-cluster-ccs-analysis
Author	Alhassan Abdulai
Institution	Central European University & University of Manchester
Language	Python 3.10+
Licence	MIT
Deposited	May 2026

Script Inventory

Table A7 Analysis Script Inventory for Analysis Replication

Script	Objective	Description
01_tea_monte_carlo.py	2	Levelised cost of CO2 avoided (LCO2A) and Monte Carlo sensitivity (N = 10,000)
02_mac_opportunity_cost.py	2	Marginal abatement cost curve and opportunity-cost analysis (nine pathways)
03_policy_lockin_scoring.py	1	Policy corpus lock-in scoring following Erickson et al. (2015)
04_slo_discourse_analysis.py	3	SLO continuum distribution, frame families, and VADER cross-tabulation
05_ets_subsidy_breakeven.py	2	UK ETS civil-penalty price series and 15-year subsidy breakeven calculations
06_joint_display_integration.py	Integration	Mixed-methods joint display figure (Figure 4.4.1)

Citation: Abdulai, A. (2026) Carbon Capture and Storage in the UK Cement Sector: Policy Incentives, Economic Viability and Public Perception. MSc thesis. Central European University & University of Manchester. Code repository available at: https://github.com/Authenticscholar/peak-cluster-ccs-analysis

APPENDIX H – PUBLIC OPINION CORPUS STATISTICS

Table A8 Public opinion corpus summary statistics

Dimension	Category	Count	Share (%)
Source type	Facebook posts	132	67%
Source type	Commentary / campaign text	35	18%
Source type	Social / post capture	28	14%
Source type	News article	2	1%
Stance	Oppose	95	48%
Stance	Neutral / mixed	80	41%
Stance	Critical scrutiny	19	10%
Stance	Support (proponent)	2	1%
Stance	Conditional / engaged	1	0.5%
Sentiment	Positive (VADER)	144	73%
Sentiment	Negative (VADER)	45	23%
Sentiment	Neutral (VADER)	8	4%
Geography	Multi-route / unspecified	69	35%
Geography	Wirral	36	18%
Geography	Cheshire	14	7%
Geography	Wirral & High Peak/Derbyshire	13	7%
Geography	High Peak / Derbyshire	13	7%
Geography	Other multi-location combinations	52	26%

APPENDIX I – SUBSIDY BREAKEVEN MODEL ASSUMPTION NOTES

Table A9 Subsidy breakeven results for CCS (post-combustion MEA)

Metric	ETS Low (USD 65/t)	ETS Central (USD 119/t)	ETS High (USD 185/t)	Throughput (Mt/yr)	Contract Term
Median LCO_2A (USD/t)	131.4	131.4	131.4	3.0	15 years
Subsidy per tonne (USD/t)	66.2	12.6	0.0	—	—
Annual subsidy (USD m/yr)	198.6	37.8	0.0	—	—
15-yr total subsidy (USD m)	**2,979**	**567**	**0.0**	—	—

Computed under three UK ETS price scenarios. Values in 2024–25 USD. 15-year subsidy undiscounted. The ETS High scenario represents the carbon price at which MEA CCS becomes marginally viable without subsidy (ETS price ≥ LCO_2A median of USD 131.4/t).

Note: Under the ETS Low scenario (current 2026 carbon price), public subsidy support for the Peak Cluster MEA configuration amounts to approximately USD 199 million per year or USD 2.98 billion over the ICC contract term (undiscounted).